

Derivational Economy in Composite Probing: The Case of The *lian...dou* Construction in Mandarin Chinese

Huacheng Cao

Department of Linguistics, University College London, London, UK.

Corresponding author(s). E-mail(s): huacheng.cao.19@ucl.ac.uk;

Abstract

Since van Urk (2015), composite $[A/\bar{A}]$ probes have been widely adopted for movements that exhibit both A- and $\bar{A}$ -properties. However, the interaction between the $[A]$ and $[\bar{A}]$ features in composite probes is still poorly understood, particularly regarding which feature acts as the movement trigger. This paper addresses this gap by investigating the *lian...dou* construction in Mandarin Chinese. Although movements in the *lian...dou* construction are motivated by the same focus interpretation, DP foci undergo A-movement whereas VP and CP foci undergo $\bar{A}$ -movement. I argue that this state of affairs follows from the composite $[A/\bar{A}]$ probe involved in the *lian...dou* construction and its interaction with principles of economy. VP and CP foci agree with the $[A/\bar{A}]$ probe only in the $[\bar{A}]$ feature, resulting in $\bar{A}$ -movement of these foci. By contrast, DP foci are fully matching goals. When both types of movement are possible for DP foci, an explicit economy calculus implies that A-movement is a more economical derivation than $\bar{A}$ -movement. Consequently, while DP foci undergo A-movement due to economy, only $\bar{A}$ -movement is available for VP/ CP foci.

Keywords: the *lian...dou* construction, Mandarin Chinese, focus movement, the A/ $\bar{A}$ distinction, composite probes, economy

1 Introduction

One of the key discoveries in linguistic theory is the identification of two distinct types of phrasal movement: A- and $\bar{A}$ -movement (Postal, 1971; Chomsky, 1977, 1981 and subsequent works). Table 1 summarises some of the mutually exclusive properties that distinguish A- and $\bar{A}$ -movement, also known as the A- and $\bar{A}$ -properties (for recent overviews on distinct properties of A- and $\bar{A}$ -movement, see van Urk, 2015; Safir, 2019; Lohninger, 2025a).

Table 1 A- and $\bar{A}$ -properties in movement operations

Properties	A-movement	$\bar{A}$ -movement
Binds an anaphor	Yes	No
Is the subject of a depictive	Yes	No
Licenses parasitic gaps	No	Yes
Induces weak crossover (WCO) effect	No	Yes
Reconstructs for binding and scope	Limited	Mostly yes
Can bypass intervening arguments	No	Yes
Is restricted to argument	Yes	No
Can induce ϕ -agreement	Yes	No
Yields information structural effects	No	Yes

Since the GB era, the A/ $\bar{A}$ distinction has been linked to properties of the landing sites, which are classified into inherent A- and $\bar{A}$ -positions (Chomsky, 1981; Déprez, 1989; Mahajan, 1990; Williams, 2003). More recently, there has been a tendency to define the A/ $\bar{A}$ distinction in terms of the agreeing feature(s) present on the probing head (Chomsky, 2007; Obata and Epstein, 2011; Abels, 2012; van Urk, 2015). [A] features (e.g., [ϕ], [case], [D]) trigger A-movement, while [$\bar{A}$] features (e.g., [wh], [top]) trigger $\bar{A}$ -movement.

The feature-based approach has received empirical support from movement configurations that simultaneously exhibit A- and $\bar{A}$ -properties, known as composite A/ $\bar{A}$ movement. Assuming that multiple agreeing features on a single head can act in unison, composite A/ $\bar{A}$ movement has been subsumed under analyses involving composite [A/ $\bar{A}$] probes (van Urk, 2015; Longenbaugh, 2017; Erlewine, 2018; Coon and Keine, 2021; Scott, 2021; Lohninger et al., 2022; Chen, 2023; Branan and Erlewine, 2024; Lohninger, 2025a). For example, in Dinka, Topicalisation of an object (an $\bar{A}$ -property) is associated with ϕ -agreement (an A-property) at the CP level, as notated by the underlines:

- (1) (van Urk, 2015, p. 103)
Miir à-càa tĩng.
giraffe 3SG-PRF.1SG see.NF
‘A giraffe, I have seen.’

Based on a survey of 14 languages across 9 language families, Lohninger (2025a) shows that composite A/ $\bar{A}$ constructions do not show random combinations of A- and $\bar{A}$ -properties. Composite A/ $\bar{A}$ constructions may exhibit both A- and $\bar{A}$ -properties that are shaded grey in Tables 1 (i.e., with positive values for these properties, such

as the Dinka example in (1)). In contrast, for properties that are not shaded grey, composite $A/\bar{A}$ constructions parallel typical A- or $\bar{A}$ -movement. Lohninger (2025a) generalises that, even though $[A]$ and $[\bar{A}]$ features may form a composite $[A/\bar{A}]$ probe, the movement chain it triggers is either an A- or $\bar{A}$ -chain, but not a chain with mixed A- and $\bar{A}$ -properties. Although Lohninger’s (2025a) generalisation enhances our understanding of the nature of movement chains in composite $A/\bar{A}$ movement, it remains unclear what determines whether a composite $[A/\bar{A}]$ probe triggers A- or $\bar{A}$ -movement.

The *lian...dou* construction in Mandarin Chinese represents a crucial typological case that sheds light on this question. On the one hand, movements in the *lian...dou* construction align with Lohninger’s (2025a) generalisation that the movement chain is either an A- or $\bar{A}$ -chain. On the other hand, this construction displays an intriguing pattern where the type of movement alternates with the syntactic category being fronted. Specifically, with the constant focus interpretation across fronted categories, DP foci undergo A-movement while VP/CP foci undergo $\bar{A}$ -movement. For illustration, consider the following examples.

- (2) Wo *lian* Zhangsan_k *dou* *cong* ta-ziji_k *de* jia li ganzou le t_k.
 I *lian* Zhangsan *dou* from s/he-self POSS home in dislodge ASP
 ‘I dislodged even Zhangsan_k from his_k own house.’

In example (2), the movement of the focus *Zhangsan* yields a focus interpretation comparable to the meaning of English *even*. Meanwhile, the fronted DP binds the anaphor *taziji* at the landing site, a typical property of A-movement.¹

As for VPs and CPs, the consensus in the literature is that they generally do not undergo A-movement. Although most tests for A- and $\bar{A}$ -properties cannot be applied to VP and CP foci, I will show that DP and VP/CP foci do behave differently in reconstruction effects. In example (3), the anaphor *taziji* within the fronted DP cannot be bound under reconstruction. By contrast, when the anaphor is embedded in the fronted VP in (4) and the fronted CP in (5), it is bound by a structurally lower argument, namely, *Zhangsan*.² Given that A-movement of DP foci does not reconstruct, I take it as evidence that VP and CP foci undergo $\bar{A}$ -movement.

- (3) Wo *lian* [_{DP} ta-ziji_k *de* fangzi] *dou* bi Zhangsan_k maidiao le.
 I *lian* [_{DP} s/he-self POSS house] *dou* force Zhangsan sell ASP
 Intended: ‘I forced Zhangsan_k to sell even his_k own house.’
- (4) Wo *lian* [_{VP} maidiao ta-ziji_k *de* fangzi] *dou* bi Zhangsan_k zuo le.
 I *lian* [_{VP} sell s/he-self POSS house] *dou* force Zhangsan do ASP
 ‘I forced Zhangsan_k even to sell his_k own house.’

¹All judgments are either taken from the literature or my consultants. The core observation has been verified by 17 native speakers from mainland China. Their judgments are unanimous. Some reconstruction readings are slightly hard to access without context. Due to limited space, I will provide English translations of the contexts.

²I follow Shyu (1995) in assuming that *zuo* ‘do’ is a dummy verb, similar to supportive *do* in English.

- (5) Wo *lian* [_{CP} *ta*-*ziji*_k *de* *erzi* *sheng-bing* *le*] *dou* *mei* *gaosu* *Zhangsan*_k.
 I *lian* [_{CP} *s/he*-self POSS son get-sick SFP] *dou* NEG tell *Zhangsan*
 ‘I did not even tell Zhangsan_k that his_k own son got sick.’

I propose that the *lian...dou* construction involves a composite $[A/\bar{A}]$ probe, consisting of a $[\phi]$ feature and an $[\text{EVEN}]$ feature that encodes the focus interpretation (see also Chen, 2023). VP and CP foci agree with the $[\phi/\text{EVEN}]$ probe only in the $[\text{EVEN}]$ feature, resulting in $\bar{A}$ -movement of these foci. On the other hand, DP foci match with the $[\phi / \text{EVEN}]$ probe in both $[\phi]$ and $[\text{EVEN}]$ features. In principle, the two agreed features can trigger A- or $\bar{A}$ -movement of DP foci. However, A-movement blocks $\bar{A}$ -movement because the former is the more economical derivation.

Under the copy theory of movement (Chomsky, 1993, 1995), movement chains are interpreted by distinct interpretation rules: A-chains abstract over individual variables, whereas $\bar{A}$ -chains abstract over choice function variables (Sauerland, 1998, 2004; Ruys, 2000; Poole, 2024; Poole and Keine, 2024). I will show that the interpretation rule for A-chains involves fewer derivational steps than that for $\bar{A}$ -chains. Consequently, while A- and $\bar{A}$ -chains yield the same focus interpretation of DP foci, economy prefers the former.

The argumentation proceeds as follows. Sect. 2 provides an overview of the *lian...dou* construction. I demonstrate that all categories move to the same position and for the same focus interpretation. In Sect. 3, I show the A- and $\bar{A}$ -properties of DP foci and argue that these properties result from the properties of the $[\phi/\text{EVEN}]$ probe involved. Sect. 4. shows that in contrast to DP foci, VP and CP foci undergo $\bar{A}$ -movement. In section 5, I point out the main concern of this paper: what makes the triggering feature different in cases of DP and non-DP foci? Sect. 6 develops an economy-based account. Sect. 7 concludes.

2 An overview of the *lian...dou* construction

In Mandarin Chinese, two particles are used to express the meaning of *even*: *dou* and *lian*, as in (6).³ *Dou* is obligatory, occupying a preverbal position. *Lian* is optional. It combines with a focused phrase that bears the *even*-meaning. The combination of *lian* and the focused phrase (the *lian*-phrase) must c-command *dou*. As shown by the contrast between examples (7a and b), the focused object undergoes phrasal movement to a position in between the subject and *dou* (Shyu, 1995; Badan, 2008; Constant and Gu, 2010; Badan and Del Gobbo, 2015).⁴

³The additive particle *ye* ‘also’ can replace *dou* and co-occurs with *lian*-phrases. However, using *ye* does not change the A- or $\bar{A}$ -properties of fronted categories. The choice between them is largely determined by pragmatic effects (Badan, 2012).

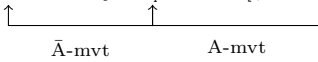
⁴Fronted foci can occur in a TP-internal position or a TP-external position (Shyu, 1995; Badan, 2008; Constant and Gu, 2010; Badan and Del Gobbo, 2015; Chen, 2023). Unlike TP-internal *lian*-DPs, TP-external *lian*-DPs show pure $\bar{A}$ -properties. First, examples (i) and (ii) show that a TP-external *lian*-DP can take a topic maker *a* or a resumptive pronoun *ta*, but a TP-internal *lian*-DP is incompatible with either. Second, unlike the TP-internal *lian*-DP shown in example (5), a TP-external *lian*-DP reconstructs for anaphor binding, as in (iii). The derivation of TP-external *lian*-DPs is schematically shown in (iv).

- (i) Zhangsan *lian* [*Li jiaoshou*]_i (**a*) *dou* *bu* *xinren* (**ta*)_i.
 Zhangsan *lian* [*Li professor*] TOP *dou* NEG trust *s/he*
 ‘Zhangsan does not trust even Professor Li.’

- (6) (lian) Zhangsan *(dou) chi-dao le.
 lian Zhangsan dou late-arrive SFP
 ‘Even Zhangsan arrived late.’
- (7) a. *Zhāngsan dou gan chi lian chongzi.
 Zhangsan dou dare eat lian insect
 b. Zhangsan lian chongzi_i dou gan chi t_i.
 Zhangsan lian insect dou dare eat
 ‘Zhangsan dares to eat even insects.’

As noted by Shyu (1995), VP and CP foci also move to a TP-internal position, as shown in (8) and (9).⁵

- (8) Zhangsan lian [VP chi chongzi]_i dou gan zuo t_i.
 Zhangsan lian [VP eat insect]_i dou dare do
 ‘Zhangsan even dares to eat insects.’

-
- (ii) lian [Li jiaoshou]_i (a) Zhangsan dou bu xinren (ta)_i.
 lian [Li professor]_i TOP Zhangsan dou NEG trust s/he
 ‘Even Professor Li, Zhangsan does not trust (him/her).’
- (iii) lian [ta-ziji_{k/*m}] de pengyou_i Zhangsan_k dou bi Lisi_m ma guo t_i.
 lian [s/he-self POSS friend]_i Zhangsan dou force Lisi scold EXP
 ‘Even his_{k/*m} own friends, Zhangsan_k forced Lisi_m to scold (him_{k/*m}).’
- (iv) Lian-DP_i Subject t’_i dou ... [VP ... t_i ...]


Following the researchers cited above, I assume that TP-external *lian*-phrases are derived by further topicalisation of TP-internal *lian*-phrases, which is an independent $\bar{A}$ -operation in Mandarin Chinese. This paper focuses on internal *lian*-phrases because internal *lian*-DPs show composite A/ $\bar{A}$ -properties.

⁵Cheng and Vicente (2013) argue that examples like (i) are derived by head movement, with the peculiar property of spelling out more than one copy. However, the head movement analysis is untenable. First, the VPs in (ii) are not full-fledged copies of each other since they take distinct objects. Second, Sect. 4.1 shows that movement of VPs obligatorily reconstructs for binding and scope. If VP doubling were derived by movement, it is surprising that the higher VP does not reconstruct, as shown by the contrast between examples (iii) and (iv). Therefore, I follow Hsu (2008, 2013) and Yin (2023) in arguing that in these examples, an adjunct VP focus is base-generated above *dou*.

- (i) Ta lian kan dou bu kan.
 s/he lian see dou NEG see
 ‘S/he is even not willing to take a look.’
- (ii) Ta lian he shui dou zhi he Bali-shui.
 s/he lian drink water dou only drink Perrier-water
 ‘S/he drinks only Perrier even when s/he just needs to drink some water’
- (iii) Lisi lian [VP nian mou-ge xuesheng de mingzi] dou you liang-ci [VP nian-cuo le].
 Lisi lian [VP spell some-CLF student POSS mingzi] dou have two-CLF [VP spell-wrong ASP]
 some > two: ‘For some students, Lisi even mispronounced their names twice.’
 *two > some: ‘There are two occasions on which Lisi even mispronounced the names of some (different) students.’
- (iv) Lisi you liang-ci lian [VP nian mou-ge xuesheng de mingzi] dou [VP nian-cuo le].
 Lisi have two-CLF lian [VP spell some-CLF student POSS mingzi] dou [VP spell-wrong ASP]
 *some > two: ‘For some students, Lisi even mispronounced their names twice.’
 two > some: ‘There are two occasions on which Lisi even mispronounced the names of some (different) students.’

- (9) Zhangsan *lian* [_{CP} *diqu shi yuan-de*]_i *dou bu xiangxin* *t_i*.
 Zhangsan *lian* [_{CP} earth COP round-ADJ] *dou* NEG believe
 ‘Zhangsan does not even believe that the Earth is round.’

2.1 The same landing site across all fronted categories

Preverbal adverbials in Mandarin Chinese obey the same constraint on the order of adverbials as do all other languages with a well-studied adverbial system, as schematised in (10) (Cinque, 1999; Ernst, 2014a,b).

- (10) Speaker-orientated » Subject-orientated » Neg » Manner/degree

In the *lian...dou* construction, all fronted categories show the same rigid order with respect to preverbal adverbials. Specifically, the cluster ‘*lian*-phrase *dou*’ must be higher than negation but lower subject-orientated adverbs.⁶ Examples (11)-(13) show that negation cannot precede *lian*-phrases across all fronted categories.

- (11) Zhangsan <*meiyou> *lian* [_{DP} *zhe ben shu*]_i *dou* <meiyou> *kan* *t_i*.
 Zhangsan NEG *lian* [_{DP} this CLF book] *dou* NEG read
 ‘Zhangsan didn’t read even this book.’
- (12) Zhangsant_k <*meiyou> *lian* [_{VP} *xi yifu*]_i *dou* <meiyou> *bang* *ta_k de*
 Zhangsan NEG *lian* [_{VP} wash cloth] *dou* NEG help he POSS
haizi *zuo guo* *t_i*.
 kid do EXP
 ‘Zhangsan didn’t even do laundry for his children.’
- (13) Zhangsant_k <*meiyou> *lian* [_{CP} *ta_k tou guo dongxi*]_i *dou* <meiyou>
 Zhangsan NEG *lian* [_{CP} he steal EXP thing] *dou* NEG
chengren *t_i*.
 admit
 ‘Zhangsan_k didn’t even admit that he_k had stolen anything.’

The adverb *congming-de* is ambiguous between a manner and a subject-oriented interpretation. In examples (14) and (15), the adverb *congming-de* has the subject-orientated interpretation when this adverb precedes *lian*. By contrast, the same adverb only has the manner interpretation when it follows *dou*.

- (14) Zhangsan <hen congming-de> *lian* [_{DP} *jizhe zui nan-de*] *wenti*
 Zhangsan very smart-ADV *lian* [_{DP} reporter most difficult-ADJ question
]_i *dou* <hen congming-de> *huibi* *le* *t_i*.
] *dou* very smart-ADV sidestep ASP
congming-de > *lian*: ‘Zhangsan is smart in that he sidestepped even the most

⁶Adverbials cannot be attached in between IP internal *lian*-phrases and *dou*. As shown in (i), while the adverb *jingchang* ‘often’ can adjoin higher than the *lian*-phrase or lower than *dou*, it cannot adjoin immediately above *dou*. The reader may refer to Cao (2025) for a semantic-based account for this adjacency effect.

- (i) Zhangsan <jingchang> *lian* *fan* <*jingchang> *dou* <jingchang> *bu* *chi*.
 Zhangsan often *lian* meal often *dou* often NEG eat
 ‘Zhangsan often doesn’t even eat.’

difficult question from the reporter.

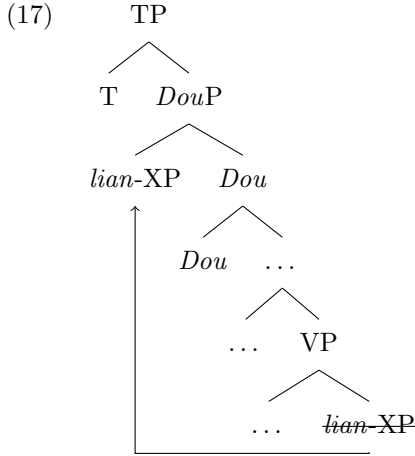
dou > *congming-de*: ‘Zhangsan sidestepped even the most difficult question from the reporter in a smart way.’

- (15) Zhangsan <hen *congming-de*> *lian* [_{VP} *hui*bi *jizhe* *zui* *nan-de* Zhangsan very smart-ADV *lian* [_{VP} sidestep reporter most difficult-ADJ *wenti*]_i *dou* <hen *congming-de*> bang Lisi *zuo* le *t_i*.
question] *dou* very smart-ADV help Lisi do ASP
congming-de > *lian*: ‘Zhangsan is smart in that he helped Lisi sidestep even the most difficult question from the reporter.’
dou > *congming-de*: ‘Zhangsan helped Lisi sidestep even the most difficult question from the reporter in a smart way.’

Example (16) involves a fronted CP and another ambiguous adverb *laoshi-de*. Similar to what we have seen for DP and VP cases, this adverb is disambiguated depending on whether it precedes *lian* or follows *dou*.

- (16) Zhangsan_k <hen *laoshi-de*> *lian* [_{CP} *ta_k* *tou* *guo* *dongxi*]_i *dou* <hen Zhangsan very honest-ADV *lian* [_{CP} he steal EXP thing] *dou* very *laoshi-de*> *chengren* le *t_i*.
honest-ADV admit ASP.
laoshi-de > *lian*: ‘Zhangsan is honest in that he even committed theft.’
dou > *laoshi-de*: ‘Zhangsan even committed theft in an honest way.’

Assuming that subject-orientated adverbs attach to TP and negation sits higher than the extended verbal projection, these examples suggest that foci of different categories uniformly move to a designated position in between TP and the extended verbal projection. In line with previous analyses (Qu, 1994; Shyu, 1995, 2001; Paul, 2002; Badan, 2008; Constant and Gu, 2010), I argue that *dou* heads a projection *DouP* in the TP-internal domain, and that movement of *lian*-phrases (*lian*-XP) targets the specifier of *DouP*, as depicted in (17).



2.2 The same focus interpretation across all fronted categories

In English, the focus interpretation of *even* is to introduce an additive presupposition and a scalar presupposition (Karttunen and Peters, 1979; Rooth, 1985; Krifka, 1991; Crnič, 2011). Under Roothian focus semantics (Rooth, 1985, 1992), the additive presupposition asserts that an alternative proposition distinct from the prejacent (the semantics of the sentence without focus particles) is true. The scalar presupposition asserts that for every true alternative proposition, the likelihood of that proposition exceeds the likelihood of the prejacent. Throughout the paper, I refer to the focus interpretation of *even* as the EVEN interpretation.

It is generally agreed upon that the *lian...dou* construction expresses the same EVEN interpretation as *even* in English (Shyu, 1995, 2018; Badan, 2008; Xiang, 2008; Liao, 2011; Xiang, 2020). For concreteness, I follow Shyu (1995) in assuming that *lian* is a scalar particle and *dou* is a focus-sensitive quantifier. The exact meanings of *lian* and *dou* are orthogonal to the focus of this paper. What is crucial is that, in the *lian...dou* construction, movement of *lian*-phrases is uniformly motivated by the EVEN interpretation.

In examples (18)-(20), the scalar particle *lian* combines with different categories. However, the *lian*-phrase in each case contains the same focused expression, namely, *laoban* ‘boss’.⁷ The focus interpretation in each example is also roughly the same: compared to criticising others, *Zhangsan* is less likely to criticise the boss. This suggests that as long as the alternative-generating focus remains the same, the focus interpretation is constant across all fronted categories. In other words, differences in syntactic categories do not affect the focus interpretation.

- (18) Zhangsan *lian* [_{DP} dui laoban_F de pippng]_i dou gan dang-mian
 Zhangsan *lian* [_{DP} against boss COMP criticism] dou dare at-face
 ti-chulai t_i.
 put-forward
 ‘Zhangsan dares to criticise even the boss_F face to face.’
- (19) Zhangsan *lian* [_{VP} dangmian piping laoban_F]_i dou gan zuo t_i.
 Zhangsan *lian* [_{VP} at-face criticise boss] dou dare do t
 ‘Zhangsan even dares to criticise the boss_F face to face.’
- (20) Lisi *lian* [_{CP} Zhangsan dangmian piping le laoban_F]_i dou bu zhidao t_i.
 Lisi *lian* [_{CP} Zhangsan at-face criticise ASP boss] dou NEG know
 ‘Lisi did not even know that Zhangsan criticised the boss_F face to face.’

⁷I use ‘___F’ to indicate the alternative-generating focus. For simplicity, I describe fronted categories as focused phrases. However, what is fronted is essentially the category to which the squiggle operator attaches. It can be the focus or contain the focus. This complication has no bearing on the argumentation. Constant and Gu (2010) argue that movement in the *lian...dou* construction can target a subpart of the focus (see also Li, 2020). They reach this conclusion based on the assumption that there is only one focus per sentence. However, as Neeleman and Szendrői (2004) show, in a sentence, a focus can be embedded in a larger focus. In example (i), the fronted DP focus expresses the EVEN interpretation. At the same time, this DP focus is embedded in VP₁, which serves as a larger contrastive focus. Therefore, being a subpart of a larger focus does not invalidate the focus nature of *lian*-phrases.

(i) Zhangsan_k ningyuan [_{VP1} *lian* [_{DP} zijik de xingming] dou dashang], ye juebu [_{VP2} touxiang].
 Zhangsan would [_{VP1} *lian* [_{DP} self_k POSS life] dou give-up], also NEG [_{VP2} surrender]
 ‘Zhangsan would rather give up even his own life, rather than surrender.’

Sentences are uniformly infelicitous if foci are construed as at the higher end of a likelihood scale, regardless of their syntactic categories. In examples (21)-(23), the focused phrases introduce a scale of water and its alternatives. However, under the context described in (21), water is unlikely to be ranked lower on a likelihood scale.

- (21) *Context: people must keep themselves hydrated, particularly during hot days.*
 #Lisi lian [_{DP} shui_F]_i dou jingchang he t_i.
 Lisi lian [_{DP} water]_i dou always drink
 ‘Lisi always drinks even water_F.’
- (22) #Lisi lian [_{VP} he shui_F]_i dou jingchang zuo t_i.
 Lisi lian [_{VP} drink water]_i dou always do
 ‘Lisi always even drinks water_F.’
- (23) #Lisi lian [_{CP} ren xuyao he shui_F]_i dou zhidao t_i.
 Lisi lian [_{CP} people need drink water]_i dou know
 ‘Lisi even knows that people need to drink water_F.’

3 The A/ $\bar{A}$ character of the *lian... dou* construction

Although the landing site and the EVEN interpretation are the same across fronted categories, DP and non-DP foci undergo movement of different types. As summarised in Table 2, movement of DPs shows composite A/ $\bar{A}$ properties. On the one hand, fronted DPs exhibit typical A-properties at the landing site. On the other hand, movement of DPs resembles $\bar{A}$ -movement in that it can bypass intervening arguments and is triggered by information structural effects. In contrast, movement of VPs and CPs shows pure $\bar{A}$ -properties. In this section, I will discuss A- and $\bar{A}$ -properties of fronted DPs and argue that these properties follow from the properties of the composite [ϕ / EVEN] probe. $\bar{A}$ -properties of fronted VPs and CPs will be postponed to Section 4.⁸

⁸In the *lian... dou* construction, movement types exhibit the typical A/ $\bar{A}$ distinction with respect to the locality of movement. While local movement of DPs shows A-properties, long-distance movement of DP and non-DP foci uniformly involves $\bar{A}$ -movement. Due to limited space, I present examples of fronted DPs and VPs in (i) and (ii). In contrast to local A-movement in example (3), long-distance movement of the DP in (i) reconstructs for anaphor binding. In example (ii), the binding possibility suggests that the fronted VP reconstructs to its base position, a typical property of $\bar{A}$ -moved VPs.

- (i) <lian [_{DP} ta-ziji_k/*_m de pengyou]_i> Zhangsan_k <*lian [_{DP} ta-ziji_k de pengyou]_i> dou juede
 lian [_{DP} s/he-self POSS friend]_i Zhangsan lian [_{DP} s/he-self POSS friend]_i dou juede
 [_{CP} Lisi_m ma guo t_i].
 [_{CP} Lisi scold EXP]_i
 ‘Even his_k/*_m own friends, Zhangsan_k believes that Lisi_m scolded (them).’
- (ii) <lian [_{VP} ma ta-ziji_k/*_m de pengyou]_i> Zhangsan_k <*lian [_{VP} ma ta-ziji_k/*_m de pengyou]_i>
 lian [_{VP} scold s/he-self POSS friend]_i Zhangsan lian [_{VP} scold s/he-self POSS friend]_i
]> dou juede [_{CP} Lisi_m zuo guo t_i].
] dou think [_{CP} Lisi do EXP]_i
 ‘Even scold his_k/*_m own friends, Zhangsan_k believes that Lisi_m did.’

However, an additional complication arises from the landing site of long-distance movement. Long-distance movement of all categories cannot target a TP-internal position; it must instead target the left periphery. I attribute this restriction to an independent constraint on movement paths, known as the Generalised Ban on Improper Movement (van Riemsdijk and Williams, 1981; Williams, 2003; Abels, 2007; Neeleman and van de Koot, 2010; Keine, 2019):

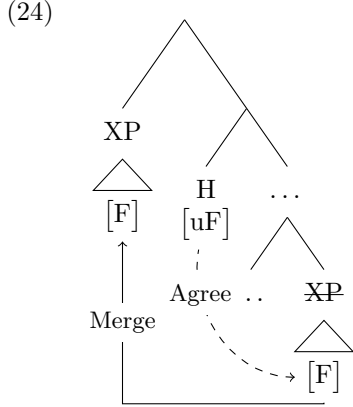
- (iii) Generalised Ban on Improper Movement (GBOIM, adopted from Poole, 2023, p. 238)
 Movement to [Spec, XP] cannot proceed from [Spec, YP] or across YP, where Y is higher than X in the functional sequence (e.g., [_{CP} ... [_{TP} ... [_{VP} ...]]]).

Table 2 The A/ $\bar{A}$ character of the *lian... dou* construction

	DPs	VP/CPs
Yields information structural effects	Yes	Yes
Can bypass intervening arguments	Yes	Yes
Reconstructs for binding and scope	No	Yes
Binds an anaphor	Yes	Not applicable (NA)
Is the subject of a depictive	Yes	NA
Induces weak crossover (WCO) effect	No	NA

3.1 Backgrounds on composite A/ $\bar{A}$ movement

A major claim in minimalist syntax is that all instances of phrasal movement are the output of Merge, accompanied by a process of feature checking (Chomsky, 1995, 2001). In this paper, I adopt a specific implementation that all phrasal movement results from the successive application of Agree and Merge. The dashed line in (24) illustrates Agree: a head H with an unvalued feature $[uF]$ (the probe) searches for the closest phrase XP with a matching valued feature $[F]$ (the goal). Then the probe attracts the goal to the specifier of the probing head, as indicated by the solid line in (24).



Under the featural approach to movement (Chomsky, 2007; Obata and Epstein, 2011; Abels, 2012; van Urk, 2015), movement types are solely derived from properties of the feature(s) involved in Agree. Agreeing features are classified into $[A]$ features, such as $[D]$ and $[\phi]$ features, and $[\bar{A}]$ features, such as $[wh]$, $[top]$ and $[foc]$ features. Instances of the former trigger A-movement, and instances of the latter trigger $\bar{A}$ -movement.

One characteristic of the featural (but not of the positional) approach is that $[A]$ and $[\bar{A}]$ features can form a composite $[A/\bar{A}]$ probe, searching for a goal with the same set of features (Aldridge, 2004; Coon and Bale, 2014; van Urk, 2015; Bossi and Diercks, 2019; Coon and Keine, 2021; Scott, 2021; Branen and Erlewine, 2024). The possibility

Under GBOIM, although $\bar{A}$ -movement of *lian*-phrases can pass through CPs, it cannot target a TP-internal position in the matrix clause because TPs are structurally lower than CPs. Given this independent constraint on movement paths, a full discussion of long-distance movement falls beyond the scope of this paper. For an analysis of long-distance movement in the *lian... dou* construction, the reader may refer to Cao (2025).

of composite probes predicts that there should be composite A/ $\bar{A}$ movement where A- and $\bar{A}$ -properties co-occur in a single movement step. Since van Urk (2015), researchers have shown that composite A/ $\bar{A}$ constructions are attested in various languages (van Urk, 2015; Longenbaugh, 2017; Erlewine, 2018; Coon and Keine, 2021; Scott, 2021; Lohninger et al., 2022; Chen, 2023; Lohninger and Yip, 2023; Branan and Erlewine, 2024; Chou, 2024; Lohninger, 2025a).

Notably, composite A/ $\bar{A}$ constructions never show randomly mixed A- and $\bar{A}$ -properties. Based on a survey of 14 languages of 9 language families, Lohninger (2025a) shows that only a subset of A- and $\bar{A}$ -properties can co-occur.

Table 3 illustrates the true source of composite A/ $\bar{A}$ movement. While [A] and [$\bar{A}$] probes show mutually exclusive properties, [A/ $\bar{A}$] probes may exhibit positive values for all four properties. For example, [A/ $\bar{A}$] probes in some languages trigger movement for information structural effects and feed ϕ -agreement at the landing site. Composite A/ $\bar{A}$ movement in Dinka provides such a case. In Example (25), repeated from example (1), topicalisation of the object *Miir* (an $\bar{A}$ -property) feeds ϕ -agreement (an A-property) at the CP level, as notated by the underlines.

- (25) (van Urk, 2015, p. 103)
Miir à-càa tîing.
giraffe 3SG-PRF.1SG see.NF
‘A giraffe, I have seen.’

In contrast, movement chains never show mixed A- and $\bar{A}$ -properties. As shown in Table 4, even in composite A/ $\bar{A}$ constructions, properties that are standardly tied to A- or $\bar{A}$ -chains always pattern together in their values. In composite A/ $\bar{A}$ movement in Dinka, the resulting movement chain shows pure A-properties. In example (26), the base position of *Bòl* is below the subject *the story about himself*. The binding relation between *Bòl* and the anaphor must be determined at the landing site.

- (26) (van Urk, 2015, p. 111)
Bòl_k à-íi [àkékòol-tí è ròt-dèk_k] t_k piòolic.
Bol 3SG-PRF.OV [story-that P self-SG.3SG] criticize.NF
‘Bol_k, that story about himself_k has criticised.’

Based on these findings, Lohninger (2025a) concludes that even though A- and $\bar{A}$ -features can act as a composite probe and search for a goal in unison, the movement chain they trigger is either an A- or an $\bar{A}$ -chain, but never a mixed A/ $\bar{A}$ -chain. In other words, either the [A] feature or the [$\bar{A}$] feature in a composite [A/ $\bar{A}$] probe acts as the movement trigger, resulting in the corresponding A- or $\bar{A}$ -chain.

In the following, I will show that this cross-linguistic signature of composite A/ $\bar{A}$ movement repeats itself in movement of DP foci in the *lian...dou* construction. In a nutshell, the composite [ϕ /EVEN] probe in the *lian...dou* construction targets DPs that bear the EVEN interpretation and triggers A-movement of those DPs.⁹

⁹A position-based approach has difficulties explaining the cross-linguistic signature of composite A/ $\bar{A}$ movement. Under the position-based approach, composite A/ $\bar{A}$ movement would target a mixed A/ $\bar{A}$ -position. Since the properties of movement chains are solely determined by the nature of the landing site, a mixed A/ $\bar{A}$ -position wrongly predicts that the movement chain involved in composite A/ $\bar{A}$ movement should be a mixed A/ $\bar{A}$ -chain. For comprehensive evaluations of the position-based and feature-based approaches to composite A/ $\bar{A}$ movement, see van Urk (2015) and Lohninger (2025a).

Table 3 A/ $\bar{A}$ properties of the probing head (adapted from Lohninger, 2025a, p. 11)

Properties	[A] probe	$[\bar{A}]$ probe	[A/ $\bar{A}$] probe
Can bypass intervening arguments	No	Yes	Yes
Is restricted to argument	Yes	No	Yes
Can induce ϕ -agreement	Yes	No	Yes
Yields information structural effects	No	Yes	Yes

Table 4 A/ $\bar{A}$ properties of the movement chain (adapted from Lohninger, 2025a, p. 11)

Properties	[A] probe	$[\bar{A}]$ probe	[A/ $\bar{A}$] probe	[A/ $\bar{A}$] probe
Binds an anaphor	Yes	No	Yes	No
Is the subject of a depictive	Yes	No	Yes	No
Licenses parasitic gaps	No	Yes	No	Yes
Induces weak crossover (WCO) effect	No	Yes	No	Yes
Reconstructs for binding and scope	Limited	Mostly yes	Limited	Mostly yes

3.2 A/ $\bar{A}$ -properties of fronted DPs

3.2.1 Fronted DPs can bind an anaphor

As the launching site of *Lisi* in (27) is too low to bind the anaphor *taziji*—the binding relation in (28) must involve the landing site. This suggests that the *lian*-DP undergoes A-movement (Cao, 2020, 2024; Chen, 2023).

- (27) *Wo bi ta-ziji_k de erzi chumai le Lisi_k.
 I force s/he-self POSS son betray ASP Lisi
 Intended: ‘I forced his_k own son to betray Lisi_k.’
- (28) Wo lian Lisi_k dou bi ta-ziji_k de erzi chumai le t_k.
 I lian Lisi_k dou force s/he-self POSS son betray ASP
 ‘Even Lisi_k, I forced his_k own son to betray him_k.’

In (30), movement of the universal *meige jingli* yields a co-varying reading that the choice of office varies with the choice of manager, which is not available in (29). This indicates that the universal binds the anaphor *tamenziji* at the landing site.¹⁰

- (29) *Wo zai tamen-ziji_k de bangongshi li piping le mei-ge jingli_k
 I at they-self POSS office in criticise ASP every-CLF manager
 yi-dun.
 one-CLF
 Intended: ‘I criticized every manager_k in their_k own offices.’

¹⁰One reviewer raises the question of whether the plural anaphor *tamenziji* is co-referential with, rather than bound by, the universal quantifier. Although the anaphor *tamenziji* is morphologically plural, it must be interpreted as a bound variable in example (30). The interpretation of example (30) is that each manager was criticised once, with the criticism occurring in that manager’s own office. This co-varying reading of (30) is possible only if the plural anaphor *tamenziji* is interpreted as a bound variable, rather than referring to a group of managers.

- (30) Wo lian mei-ge jingli_k dou zai tamen-ziji_k de bangongshi li piping
 I lian every-CLF manager dou at they-self POSS office in criticise
 le t_k yi-dun.
 ASP one-CLF
 ‘I criticised even every manager_k in their_k own offices.’

3.2.2 Fronted DPs can be the subject of a depictive

Movement of DPs in the *lian...dou* construction can create novel interpretations for depictive predicates (Cao, 2020, 2024). In example (31a), the object *Mali* cannot be associated with the preverbal depictive *luo-zhe*. When the object is fronted, however, example (31b) becomes ambiguous, with *Mali* construed as the subject of the depictive.

- (31) a. Zhangsan_k luo_{k/*n}-zhe bao le Mali_n yi-xia.
 Zhangsan nude-zhe hug ASP Mary one-CLF
 ‘Zhangsan_k hugged Mary_n nude_{k/*n}.’
 b. Zhangsan_k lian Mali_n dou luo_{k/n}-zhe bao le t_n yi-xia.
 Zhangsan lian Mary dou nude-zhe hug ASP one-CLF
 ‘Zhangsan_k hugged even Mary_n nude_{k/n}.’

Although there is an ongoing debate about the character of *-zhe* (see Djamouri and Paul, 2017 among others), possible interpretations of *-zhe* predicate track the A/ $\bar{A}$ distinction in Mandarin Chinese. Example (32) is an ex-situ cleft construction in Mandarin Chinese, which involves $\bar{A}$ -dependency (see Pan, 2017 and references therein). In contrast to (31b), it cannot create a novel interpretation of *luo-zhe*. Therefore, the novel interpretation in (31b) indicates that DP foci undergo A-movement.

- (32) Shi Mali_n, Zhangsan_k luo_{k/*n}-zhe bao le t_n yixia.
 COP Mary, Zhangsan nude-zhe hug ASP one-CLF
 ‘It is Mary_n that Zhangsan_k hugged nude_{k/*n}.’

3.2.3 Fronted DPs do not induce WCO effect

In example (33), the grammaticality of the bound-variable interpretation suggests that the universal *meige haoxuesheng* can bind the pronoun *ta* at the landing site, without inducing WCO effect (Shyu, 1995; Cao, 2020, 2024; Chen, 2023).

- (33) Li lian mei-ge hao xuesheng_k dou bi ta_k de laoshi ma guo t_k.
 Li lian every-CLF good student dou force s/he POSS teacher scold EXP
 ‘Even every good student_k, Li forced their_k teacher to scold them_k.’

3.2.4 Fronted DPs do not reconstruct

Fronted DPs do not reconstruct for anaphor binding.¹¹ Although the launching site is lower than *Lihua*, the anaphor *taziji* in (34) cannot be bound by *Lihua* under

¹¹Reconstruction effects with respect to Condition C are complicated (for a discussion of the complications in Mandarin Chinese, see Wang and Liao, 2021). For space reasons, I cannot offer a detailed discussion here. Cao (2025) show that the reconstruction for Condition C in DP, VP and CP fronting in the *lian...dou* construction parallels the patterns observed in A-movement of DPs and $\bar{A}$ -movement of VPs and CPs in English. Therefore, reconstruction effects with respect to Condition C will not undermine but further support the claim that there are different movement types involved in the *lian...dou* construction.

reconstruction. This sentence is ungrammatical because the anaphor is unbound at the landing site (Shyu, 1995; Cao, 2020, 2024; Chen, 2023).

- (34) *Wo *lian* [*ta*-*ziji*_k *qian* *de* *qian*]_i *dou* *ti* *Lihua*_k *huanqing* *le* *t_i*.
 I *lian* [*s/he*-self owe REL money] *dou* for *Lihua* pay ASP
 Intended: ‘I helped *Lihua*_k pay off even the debt *he*_k owed himself_k.’

Reconstruction for scope is not possible either. Example (35) is unambiguous. The pronoun *ta* cannot be construed as a variable bound by the universal *meige mama*. Example (36) only has the surface scope reading. The reversed scope reading is unavailable, indicating that the fronted DP does not reconstruct into the scope of the universal *meige xuesheng*.

- (35) *Wo *lian* [*ta*_k *haizi* *shang-xue* *de* *feiyong*]_i *dou* *daying* *le* *mei*-*ge*
 I *lian* [*s/he* child go-school REL expense] *dou* promise ASP every-CLF
*mama*_k *chengdan* *t_i*.
 mother cover
 Intended: ‘I promised every mother_k that I will cover even the tuition fees of their_k children.’
- (36) *Li* *lian* [*yi*-*qian* *de* *jiang-jin*]_i *dou* *fengei* *le* *meige* *xuesheng* *t_i*.
Li *lian* [one-thousand ADJ bonus-cash] *dou* give ASP every student
 $\exists > \forall$: ‘*Lisi* distributed even one thousand yuan (in total) to every student.’
 $*\forall > \exists$: ‘*Zhangsan* gave each student even one thousand yuan.’

Based on the A-properties and the absence of reconstruction effects shown above, I argue that DP foci undergo A-movement.¹²

3.2.5 DP foci can bypass intervening arguments

Movement of DP foci resembles $\bar{A}$ -movement in two aspects. In Section 1.3, we have seen that movement of DP foci is motivated by the EVEN interpretation. In this section, we consider another $\bar{A}$ -property of fronted DPs. That is, movement of DPs can bypass intervening arguments.

Wh-movement in English is a typical $\bar{A}$ -movement. Example (37) shows that the embedded object *who* moves out of non-finite clauses and bypasses intervening subjects.

¹²A-movement in Mandarin Chinese generally does not reconstruct. In Mandarin Chinese, contrastive topics undergo TP-internal A-movement, without any particles (Qu, 1994; Ernst and Wang, 1995; Shyu, 2001; Sybesma, 2021). Similar to TP-internal *lian*-DPs in the *lian...dou* construction, fronted contrastive DPs do not reconstruct. In example (i), the anaphor *taziji* cannot be bound under reconstruction. In example (ii), the fronted direct object must be construed with wide scope over the universal.

- (i) (Shyu, 1995, p. 104)
 *Wo [*naxie* *ta*-*ziji*_k *de* *shu*]_i *yijing* *jiao* *Zhangsan*_k *xian* *na-zou* *le* *t_i*
 I [those *s/he*-self POSS book] already ask *Zhangsan* first take-away ASP
 ‘I have asked *Zhangsan*_k to take away his_k own books.’
- (ii) *Li* *yuanyi* [*yi*-*qian*-*kuai* *de* *jiang-jin*]_i *fengei* *meige* *xuesheng* *t_i*.
Li willing [one-thousand-yuan ADJ bonus-cash] give every student
 $\exists > \forall$: ‘*Li* is willing to distribute one thousand yuan (in total) to every student.’
 $*\forall > \exists$: ‘*Li* is willing to give each student even one thousand yuan.’

- (37) Who_i do you expect [**Mary** to persuade [**John** to meet t_i]]?


Similarly, DP foci in the *lian... dou* construction can move across non-finite clause boundaries and bypass intervening arguments, as in (38) (Shyu, 1995; Chen, 2023).

- (38) (Chen, 2024, p. 4)
 Meiyoun-*ren* *lian* [zhe-ben shu]_i dou bi [Zhang jiao [Wang kanwan t_i]].
 No-body *lian* [this-CLF book] dou force [Zhang ask [Wang read]]
 ‘Nobody forced Zhangnan to ask Wang to read even this book.’

3.3 Composite probing in the *lian... dou* construction

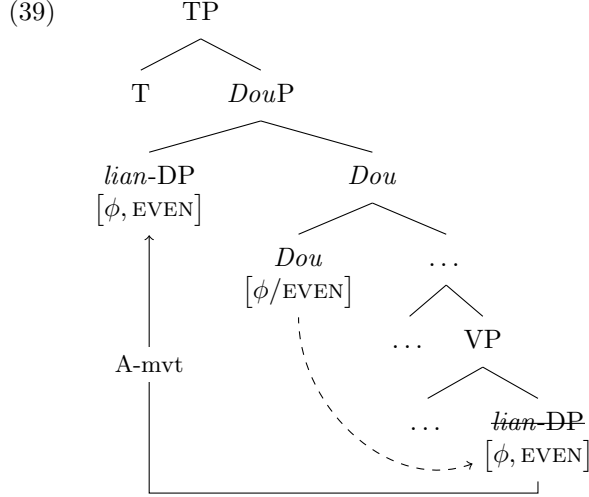
Chen (2023) takes the A/ $\bar{A}$ -properties of fronted DPs as evidence that the *lian... dou* construction involves a composite [A/ $\bar{A}$] probe. Following her approach, I argue that, in the *lian... dou* construction, the head *Dou* hosts a composite probe [ϕ /EVEN], as depicted in (39). I assume that [$\bar{A}$] features are specified for their information-structural effects, such as [wh] for *wh*-movement and [top] for topicalisation (Rizzi, 1997, 2004; Abels, 2012; van Urk, 2015). Since movement in the *lian... dou* construction consistently yields the EVEN interpretation, I assume an [EVEN] feature as the $\bar{A}$ -part of the composite probe.¹³ In line with most works in the feature-based approach to movement, I assume that A-movement is driven by [ϕ] features.¹⁴

¹³Chen (2023) follows Shyu (1995) in arguing that the information-structural feature involved is a generic [foc] feature, which forms a composite [ϕ /foc] probe with [ϕ] features. Other researchers propose that the [foc] feature should be specified for the meaning of *dou*, such as a [maximality] feature (Constant and Gu, 2010) or a [mirativity] feature (Li, 2020). I follow the latter group of researchers in assuming a specific semantic feature because not all types of foci can move above *dou*. As shown in example (i), a focus combined with the exclusive particle *zhiyou* moves above *cai*, but cannot move above *dou*. In contrast, in example (ii), foci combined with the scalar particle *lian* move above *dou*, but cannot move above *cai*. The contrast between (i) and (ii) suggests that the composite probe on *dou* must contain a specific information-structural feature, rather than a generic [foc] feature, so that *dou* probes exclusively for a *lian*-phrase.

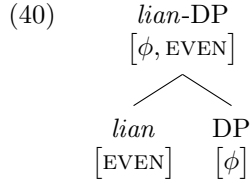
- (i) Li *zhiyou* [Qiaomusiji de shu]_i cai/*dou hui renzhen kan t_i
 Li only [Chomsky POSS book] cai/*dou will carefully read
 ‘Only Chomsky’s book, Li will read it carefully.’
 (ii) Li *lian* [Qiaomusiji de shu]_i *cai/dou bu hui renzhen kan t_i
 Li *lian* [Chomsky POSS book] *cai/dou NEG will carefully read
 ‘Even Chomsky’s book, Li will not read it carefully.’

However, there have been ongoing debates about the exact meaning of *dou* (for a recent overview, see Feng and Pan, 2023). Therefore, I use “[EVEN]” as a neutral label for the specific semantic feature involved in the *lian... dou* construction. For how the EVEN interpretation arises compositionally from the meanings of *lian* and *dou*, I will leave it for future research.

¹⁴One reviewer inquires about the origin of the [ϕ /EVEN] probe. The literature identifies various potential sources for composite [A/ $\bar{A}$] probes (see Lohninger, 2025a and references therein). One possibility is the absence of Feature Inheritance (FI) (e.g., Chomsky, 2007). FI is the hypothesis that [A] and [$\bar{A}$] features standardly enter the derivation at CPs and vPs. Subsequently, [A] features are inherited downward from C to T, and from v to V. In basic order sentences in Mandarin Chinese, the [ϕ] feature is inherited from v to V. In contrast, in the *lian... dou* construction, FI does not happen. The unvalued [ϕ] feature remains in v, forming a composite probe with the [EVEN] feature. I will leave this issue for future research.

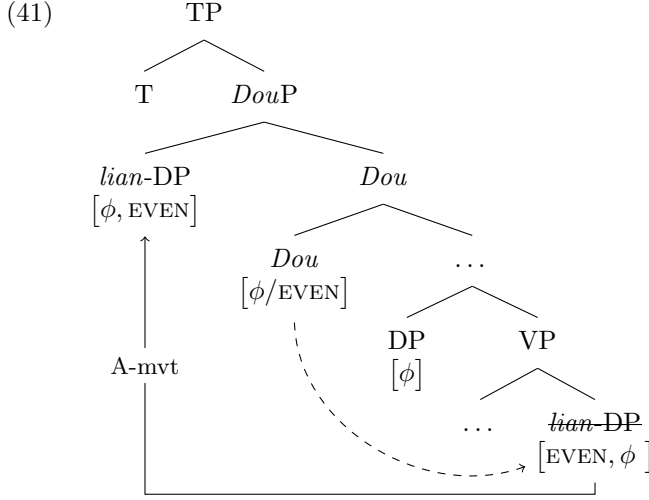


For *lian*-phrases (the goal), I assume that the scalar particle *lian* is the overt realisation of an [EVEN] feature. It adjoins to a maximal projection XP that is or contains the focus, with formal features of *lian* and its sister XP projecting to their mother (Bayer, 1996; Aoyagi, 1998; Citko, 2008; Kotek, 2014; Bárány, 2023). For example, in (40), *lian* adjoins to a DP focus. The resulting *lian*-DP projects the [EVEN] feature of *lian* and the [φ] features of the DP.



DP foci's ability to bypass intervening arguments follows from the minimality of composite probing. Minimality is a general restriction on Agree that probes must target the closest matching goal (Rizzi, 1990; Chomsky, 1995). Composite [A/ $\bar{A}$] probes differ in whether they are sensitive to a closer partial matching goal (an [A] or [$\bar{A}$] goal)(for a detailed discussion, see Lohninger, 2025a). In some languages, a closer partial matching goal blocks further probing of a fully matching goal (Erlewine, 2018; Coon and Keine, 2021; Coon et al., 2021; Scott, 2021; Lohninger et al., 2022; Branen and Erlewine, 2024). In other languages, goals that only carry a subset of the features will not terminate the probing process (van Urk, 2015; Longenbaugh, 2017; Scott, 2021; Bárány, 2023; Chen, 2023; Lohninger, 2025a). That is, a composite [A/ $\bar{A}$] probe searches for a fully matching goal and skips over partial matching goals. Chen (2023) shows that the mechanism of composite probing in the *lian*...*dou* construction belongs to the latter class. As schematically shown in (41), the [ϕ /EVEN] probe searches for a

$[\phi]$ goal that bears the EVEN interpretation. Intervening arguments with inherent $[\phi]$ features are irrelevant for the $[\phi/\text{EVEN}]$ probe.¹⁵



4 $\bar{A}$ -movement of VP and CP foci

We have seen that the A/ $\bar{A}$ -properties of DP foci straightforwardly follow from properties of the $[\phi/\text{EVEN}]$ probe. However, VP and CP foci do not show composite A/ $\bar{A}$ -properties in the way DP foci do. I will show that VP and CP foci exhibit pure $\bar{A}$ -properties. First, researchers have accepted the claim that VPs and CPs generally do not undergo A-movement but show typical $\bar{A}$ -properties. I will show that this claim also holds for fronted VPs and CPs in Mandarin Chinese. Meanwhile, in the *lian... dou* construction, in contrast to fronted DPs, fronted VPs and CPs do reconstruct. I take these as evidence that VP and CP foci undergo $\bar{A}$ -movement.

4.1 $\bar{A}$ -properties of fronted VPs

It has been widely accepted that fronted VPs generally show obligatory reconstruction to their base positions (Barss, 1986, 2001; Huang, 1993; Heycock, 1995; Takano, 1995; Sauerland and Elbourne, 2002; Kobele, 2012). Huang (1993) shows that topicalised VPs in Mandarin Chinese must reconstruct to their base positions. In example (42a), the anaphor contained in the topicalised DP can be bound by the matrix or the embedded subject. However, when the anaphor is contained in the topicalised VP, as in (42b), it must be bound by the embedded subjects.

¹⁵The minimality of composite probing predicts that if there are multiple fully matching goals, the composite probe attracts the closest one. However, it is impossible to test this prediction because multiple *lian*-phrases are not allowed within a single clause, as in (i).

(i) * Lian Zhangsan dou lian zhe-ben shu dou kan bu dong.
lian Zhangsan dou lian this-CLF book dou read NEG understand
Intended: ‘Even Zhangsan cannot understand even this book.’

- (42) (adapted from Huang, 1993, p. 119)
- a. [DP ta-ziji_{k/n} de shi]_i Zhang_k xiwang Lisi_n neng guan-yi-guan t_i.
 [DP s/he-self POSS matter] Zhang hope Lisi can care-a-little
 ‘His_{k/n} own business, Zhang_k hopes Lisi_n will care for a bit.’
- b. [VP piping ta-ziji_{*k/n}]_i Zhang_k zhidao Lisi_n juegui bu hui t_i
 [VP criticise s/he-self] Zhang know Lisi definitely not will
 ‘Criticize himself_{*k/n}, Zhang_k knows Lisi_n definitely will not.’

Fronted VPs in the *lian...dou* construction also show obligatory reconstruction for binding and scope. In (43), the anaphor *taziji* contained in the fronted VP is bound by *Lihua* under reconstruction. By contrast, when movement targets the DP, as in example (34) above, the anaphor cannot be bound.

- (43) *Context: Lihua is my most cherished son. Although he is already an adult, I still found him a suitable job and bought him a house for his wedding. But it is far from enough.*
- Wo lian [VP huanqing ta-ziji_k qian de qian]_i dou ti Lihua_k zuo le t_i.
 I lian [VP pay s/he-self owe REL money] dou for Lihua do ASP
 ‘I even helped Lihua_k pay off the debt he_k owed himself_k.’

In example (44), the pronoun *ta* is bound by the universal *meige-mama* under reconstruction. This contrasts with example (35), where movement of the DP focus does not allow the bound-variable interpretation in this context. Example (45) is unambiguous regarding its scope relation: the fronted VP must be interpreted in the scope of the universal *meige xuesheng*.

- (44) *Context: Zhangsan and I work at the Women’s Federation. We are helping each single mother find a suitable job. In order to get a promotion opportunity faster, I am willing to do more for them.*
- Wo lian [VP chengdan ta_k haizi shang-xue de feiyong]_i dou daying le
 I lian [VP cover she child go-school REL expense] dou promise ASP
 mei-ge mama_k t_i.
 every-CLF mother
 ‘I promised every mother_k that I will even cover the tuition fees of their_k children.’
- (45) Lisi_k lian [VP gei ta_k yi-bai-kuai]_i dou bi mei-ge haizi zuo t_i.
 Lisi lian [VP give s/he one-hundred-yuan] dou force every-CLF child do
 $\forall > \exists$: ‘Lisi_k forced each student even to give him_k one hundred yuan.’
 $*\exists > \forall$: ‘Lisi_k forced every child even to give him_k one hundred yuan (in total).’

4.2 $\bar{A}$ -properties of fronted CPs

Like VPs, CPs generally do not undergo A-movement. This claim is supported by observations that go back to Koster (1978). Koster (1978) shows that so-called sentential subjects actually occupy a left-peripheral topic position, rather than the canonical subject position. For example, CPs, unlike argument DPs, cannot occur in the ECM subject position:

- (46) (Alrenga, 2005, p. 185)
- a. *John believes [_{CP} that the cult members cloned a human baby] to be true.
 - b. John believes their claim to be true.

Another piece of evidence is that sentential subjects, unlike DP subjects, cannot appear after proposed auxiliaries:

- (47) (Alrenga, 2005, p. 177)
- a. *Never before has [_{CP} that you are unqualified for this job] been so obvious.
 - b. Never before has your lack of qualifications been so obvious.

In addition, in some languages, an overt DP is the real subject in sentences with so-called sentential subjects. In Dutch, the pronoun *data* occupies the subject position, with the sentence-initial CP in a left-periphery position, as in (48a). A similar English example (48b) is also acceptable when the pronoun *that* (*much*) bears sufficient emphatic stress, as notated with capital letter.

- (48) (Alrenga, 2005, p. 181)
- a. Dat hij komt (data) is duidelijk.
 hat he comes that is clear
 ‘That he will come is clear.’
 - b. That the Giants lost the World Series, THAT (MUCH) is now clear.

The same phenomenon is attested in Mandarin Chinese. In example (49), an optional DP *zhe-jian shi* follows the sentence-initial CP and functions as the subject of the verbal predicate.

- (49) [_{CP} Guo-ping dui shu le] (zhe-jian shi) rang minzhong
 [_{CP} national-pingpong team lose ASP] (this-CLF matter) make people
 wufa jieshou.
 unable accept
 ‘That the national ping pong team lost is unacceptable to the public.’

Although researchers diverge in their specific treatments of CP fronting, these facts have led them to argue that CPs can only undergo $\bar{A}$ -movement (Koster, 1978; Alrenga, 2005; Takahashi, 2010; Moulton, 2013).

Besides the general agreement that CPs cannot undergo A-movement, fronted CPs in the *lian...dou* construction behave differently from fronted DPs in that they show reconstruction effects. Recall that A-movement in Mandarin Chinese does not reconstruct, reconstruction effects in movement of CP foci further confirm its $\bar{A}$ -properties. In example (50), the embedded subject *taziji* is bound by *Zhangsan* under reconstruction.

- (50) Wo lian [_{CP} ta-ziji_k de erzi sheng-bing le] dou mei gaosu Zhangsan_k.
 I lian [_{CP} s/he-self POSS son get-sick SFP] dou NEG tell Zhangsan
 ‘I did not even tell Zhangsan_k that his_k own son got sick.’

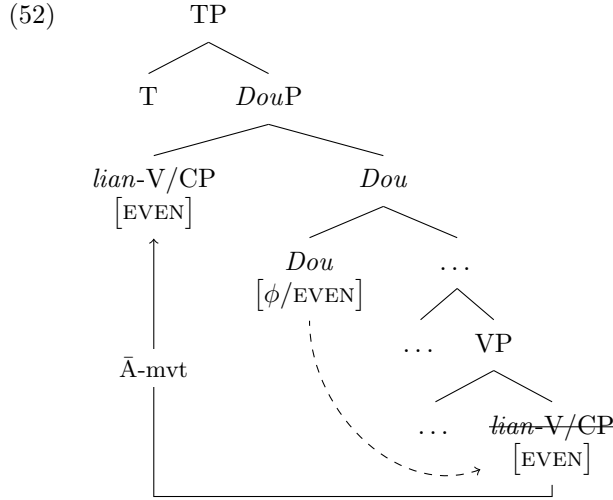
Bound-variable interpretations are also attested. The pronoun *ta* in (51) is bound under reconstruction even if the CP is fronted above the universal *meige xuesheng*.

- (51) *Context: I told the students many things about Mr. Li's misconduct. For instance, Mr. Li withheld the stipends allocated to each student and confiscated the letters they wrote to their parents. Taking the risk of being expelled,*
 Wo lian _{[CP Li laoshi cuangai le ta_k de chengji]_i} dou toutou gaosu
 I lian _{[CP Li teacher falsify ASP s/he POSS grade]_i} dou secretly tell
 le mei-ge xuesheng_k t_i.
 ASP every-CLF student
 ‘I even secretly informed every student_k that Mr. Li had falsified their_k grades.’

4.3 Independent probing in the *lian...dou* construction

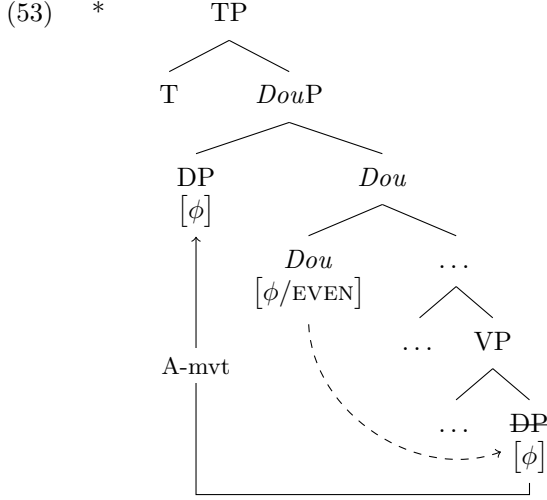
In Sect. 3, I argued that DP foci represent cases where the $[\phi/\text{EVEN}]$ probe identifies a fully matching goal. Crucially, the $[\text{EVEN}]$ feature determines that movement yields the EVEN interpretation, and the $[\phi]$ feature triggers A-movement of DP foci.

With $\bar{A}$ -properties of fronted VPs and CPs identified, we can now turn to cases in which the $[\phi/\text{EVEN}]$ probe fails to find a fully matching goal. The $\bar{A}$ -properties of VP and CP foci suggest that movement of these categories is triggered solely by the $[\text{EVEN}]$ feature, without referring to the $[\phi]$ feature. Specifically, in cases of VP/CP foci, the $[\phi/\text{EVEN}]$ probe only finds a goal that carries the $[\text{EVEN}]$ feature, as shown in (52). Since only the $[\text{EVEN}]$ part is matched between the probe and the goal, VP and CP foci undergo $\bar{A}$ -movement.



While the $[\text{EVEN}]$ feature triggers $\bar{A}$ -movement of an $[\text{EVEN}]$ goal, the $[\phi]$ feature never probes independently for a plain $[\phi]$ goal. In the following, I will show that movement of a plain $[\phi]$ goal is disallowed in the *lian...dou* construction, regardless of whether the EVEN interpretation is available (as in example (57)) or not (as in example (54)). First, movement of a $[\phi]$ goal is ungrammatical in the absence of the

EVEN interpretation. That is, the derivation depicted in (53), in which the $[\phi/\text{EVEN}]$ probe only identifies a plain $[\phi]$ goal, is not possible.



In Mandarin Chinese, DP fronting that results in an SOV order is licensed only by specific interpretive effects. The specific interpretation involved depends on prosodic stress and/or the presence of certain particles, such as the EVEN interpretation with the focus particle *dou* and contrastive readings without any particles (see footnote 12 and references therein). For example, the sentence in (54) is acceptable only when *Lisi* is stressed, marking it as the focus bearing the EVEN interpretation. In contrast, when *Lisi* is not stressed, fronting of this DP becomes unacceptable.¹⁶ I take this as evidence that the $[\phi/\text{EVEN}]$ probe cannot trigger movement of a plain $[\phi]$ goal.

- (54) You-ren Lisi dou bu renshi.
 Have-people Lisi dou NEG know
 Stress on *Lisi*: ‘Somebody does not know even Lisi.’
 *No stress on *Lisi*: ‘Somebody knows Lisi.’

Second, as depicted in (55), even if the $[\text{EVEN}]$ feature identifies an $[\text{EVEN}]$ goal, movement of a $[\phi]$ goal is still prohibited, regardless of whether the $[\phi]$ goal moves above or below the $[\text{EVEN}]$ goal. For illustration, let’s consider movement of VP foci. Examples (56) and (57) are a minimal pair. Example (56) is a control example where the $[\text{EVEN}]$ feature in the $[\phi/\text{EVEN}]$ probe triggers movement of a focused VP. In contrast, in example (57), the additional movement operation that targets the DP *Zhao* renders the sentence ungrammatical.

¹⁶In Mandarin Chinese, prosodic stress sufficiently marks an expression as the focus (Simpson, 2014; Lee, 2019 and references therein). I assume that, in examples like (54), *lian* is always present, covertly or overtly (Shyu, 1995, 2016; Kuo, 2003; Xiang, 2008).

- a. *Split*: If no such goal is found, $[A/\bar{A}]$ splits up and agrees with two different goals.
- b. *Fail*: If only one (partially fitting) goal can be found, one probe ($[\bar{A}]$ or $[A]$) can fail.

Composite probes (presumably in all languages) optimally search for a fully matching goal, referred to as *composite probing* (see Sect. 3.3 and references therein). In some languages, when no fully matching goal is available, the features bundled in composite probes can split and probe for different goals independently, a process known as *independent probing* (van Urk, 2015; Scott, 2021; Lohninger and Yip, 2023; Lohninger, 2025a). Lohninger (2025a) proposes that independent probing is a last resort to avoid a crash of the derivation. There are two options under independent probing, as shown in (58). One option is called *split* (58a), where both features probe independently and trigger separate movement operations. The other is *fail* (58b), where only one feature successfully probes and triggers movement, while the other fails to do so.

Bossi and Diercks (2019) and Scott (2021) identify languages that allow *split* while disallow *fail* (Pattern II in Table 5). Lohninger (2025a) reports two additional patterns: the languages investigated either disallow independent probing altogether (Pattern III) or permit both options (Pattern I). Based on this range of data, Lohninger (2025a) concludes that independent probing as a whole and the *fail* option under it should be parameterised across languages. Consequently, the typology of last resort probe splitting may form a hierarchy along which languages can exhibit different cut-off points: *probe together* $\gg$ *split* $\gg$ *fail*. This hierarchy implies that the availability of *fail* is dependent on *split*. That is, a language may permit *fail* only if it also permits *split*. Under this restrictive setting, Pattern IV, where *split* is not available but *fail* is allowed, is predicted to be impossible, as illustrated in the second column from the right in Table 5.

Table 5 Possible patterns in independent probing

	Independent Probing		Lohninger's (2025a) hierarchy	Independent Parameterisation
	<i>Split</i>	<i>Fail</i>		
Pattern I	Yes	Yes	✓	✓
Pattern II	Yes	No	✓	✓
Pattern III	No	No	✓	✓
Pattern IV	No	Yes	✗	✓

However, due to the limited language sample, there is an alternative parametric setting that Lohninger (2025a) does not consider. That is, the two options under independent probing—*split* and *fail*—are parameterised independently. A key difference is that, under this independent parameterisation, the availability of *fail* is no longer contingent on the availability of *split*. As a result, this alternative setting is more permissive. As indicated in the rightmost column of Table 5, it captures the attested

cross-linguistic variation and, crucially, also allows for Pattern IV. Although the language sample in Lohninger’s (2025a) study is too limited to decisively favor one setting over the other, the pattern observed in the *lian...dou* construction suggests that the more permissive setting is empirically necessary.¹⁷

In the *lian...dou* construction, *fail* is permitted, but *split* is not. The ungrammaticality of example (57) suggests that the option *split* is unavailable in the *lian...dou* construction. $\bar{A}$ -movement of VP and CP foci indicates that, if only an [EVEN] goal is available, the $[\phi]$ probe fails. Following Preminger (2009, 2011, 2014) and Georgi (2014), I assume that the $[\phi]$ probe is deleted or assigned a default value.¹⁸

The updated typology of last resort probe splitting is as follows. Some languages exhibit positive values for both *split* and *fail* (Pattern I). A second group of languages permits *split* but disallows *fail* (Pattern II). A third group disallows both *split* and *fail* (Pattern III). The *lian...dou* construction in Mandarin Chinese shows that the final logical possibility under independent parameterisation is also attested: *fail* is permitted, but *split* is not (Pattern IV).

5 The remaining question: the trigger of movement

So far, we have observed that, in the *lian...dou* construction, the $[\phi]$ feature in the $[\phi/\text{EVEN}]$ probe triggers A-movement of fully matching goals (DP foci), while the [EVEN] feature triggers $\bar{A}$ -movement of partial matching goals (VP/CP foci). Although this state of affairs fits with Lohninger’s (2025a) generalisation regarding the types of movement triggered by $[A/\bar{A}]$ probes, it raises the question of what determines whether the $[\phi/\text{EVEN}]$ probe triggers A- or $\bar{A}$ -movement? In other words, given that the [EVEN] feature can trigger $\bar{A}$ -movement of focused phrases, why does it fail to induce $\bar{A}$ -movement of DP foci?

In the next section, I will argue that although the $[\phi]$ and [EVEN] features can in principle create A- and $\bar{A}$ -chains of DP foci, economy prefers A-chains because they are less costly.

6 Derivational economy in composite probing

The distinctive characteristic of the *lian...dou* construction is that different features in the $[\phi/\text{EVEN}]$ probe act as the movement trigger, depending on whether the goal is fully matched. Specifically, while the $[\phi]$ feature triggers A-movement of fully matching goals (DP foci), the [EVEN] feature triggers $\bar{A}$ -movement of partial matching goals (VP/CP foci). I argue that the divergence in movement triggers reflects the effect of

¹⁷Lohninger (2025a) proposes that this hierarchy also reflects a ranking of probing strategies by preference. *Probe together* is the optimal strategy, followed by *split*, and finally *fail*. At first glance, the *lian...dou* construction may appear to be a counterexample. However, the main claim of this section is that *split* and *fail* should be parameterised independently. Whether universal grammar prefers one option over the other is an independent issue. One way to preserve Lohninger’s (2025a) hierarchy is to relativise it to the availability of *split* and *fail*. The revised hierarchy would be: *probe together* $\gg$ (*split*) $\gg$ (*fail*), where *split* takes precedence over *fail* only when both are part of a language’s grammar. Whether this revised hierarchy is empirically supported remains an open question, and I leave its validation to future research.

¹⁸In languages with overt ϕ -agreement, the choice between deletion and default valuation is relatively transparent, as the default value of $[\phi]$ features is generally third person singular. However, since Mandarin Chinese lacks overt ϕ -agreement, it is difficult to determine whether the $[\phi]$ probe is deleted or assigned a default value. I therefore present both possibilities in the main text and remain agnostic on this issue.

economy on the semantic derivations of movement chains. Researchers have proposed that A- and $\bar{A}$ -chains differ in their semantic derivations. Namely, A-chains abstract over individuals, whereas $\bar{A}$ -chains abstract over choice functions (Sauerland, 1998, 2004; Ruys, 2000; van Urk, 2015; Poole and Keine, 2024; Poole, 2024). I propose that the derivational costs associated with A- and $\bar{A}$ -chains are tied to the types of abstraction involved. A-chains are less costly than $\bar{A}$ -chains because abstraction over individuals requires fewer operations. Consequently, when both features are agreed in cases of DP foci, economy prefers A-chains over $\bar{A}$ -chains. I will show that the A/ $\bar{A}$ distinction in terms of reconstruction results from two distinct types of abstraction over movement chains (Sect 6.1), and that an explicit economy calculus implies that A-chains are arguably less costly than $\bar{A}$ -chains (Sect 6.2). Although economy prefers A-chains over $\bar{A}$ -chains, the interpretation rule for A-chains cannot be applied to movement of VP and CP foci (Sect 6.3).

6.1 Two types of movement chains

A- and $\bar{A}$ -movement are known to show distinct properties in terms of reconstruction. Namely, reconstruction in A-movement is more limited than that in $\bar{A}$ -movement (see Sportiche, 2017 for a recent overview). Under the copy theory of movement (Chomsky, 1993, 1995), reconstruction is accounted for by how the copies in a movement chain are modified for interpretation. Suppose that a feature [F] triggers movement of a DP headed by a semantic operator OP. The result of movement is a copy chain, as illustrated in (59).

$$(59) \quad [\text{DP OP [NP]}] \dots [\text{DP OP [NP]}]$$

Given that A-movement in Mandarin Chinese does not reconstruct, I assume that there are only two interpretation rules for movement chains, determined by the nature of the feature [F]. If feature [F] is an [A] feature, the chain in (59) will be translated into an A-chain, as formulated in (60).

- (60) Interpretation rule for A-chains (van Urk, 2015, p. 41):
In a movement structure formed by Agree for [A] features, adjoin a node λx to the probing head.

$$(61) \quad [\text{DP OP [NP]}] \lambda x_e \dots [x_e]$$

As depicted in (61), the resulting A-chain forms a predicate abstracting over individuals, with the head of the chain serving as its external argument.¹⁹ Its semantic derivation involves the application of Trace Conversion, an operation that converts the lower copy into an individual variable of type e (Sauerland, 1998, 2004; Fox, 2002; Elbourne, 2005).²⁰ As Chomsky (1995) argues, such A-chains allow argument-related properties, such as scope and binding, to be interpreted at the head of the chain, while leaving its trace to be theta-marked. In other words, A-chains do not reconstruct.²¹

¹⁹Throughout the paper, I will abstract away from intensionality and use λ -operator-variable notation as in (61) to conveniently indicate the semantic type of the variable that the lower copy will be translated into.

²⁰It has been argued that traces are complex semantic objects, namely, bound definite descriptions (Engdahl, 1980, 1986; Sauerland, 1998, 2004; Fox, 1999, 2002, 2003). However, whether traces are complex or not is orthogonal to their semantic type.

²¹In languages like English, reconstruction in A-chains is attested, but only in certain contexts (May, 1977; Barss, 1986; Hornstein, 1995; Fox, 1999; Lebeaux, 2009). Given that reconstruction in A-chains is by

In contrast, if feature $[F]$ is an $[\bar{A}]$ feature, the chain in (59) will be translated into an $\bar{A}$ -chain that abstracts over choice function variables, as stated in (62).

- (62) Interpretation rule for $\bar{A}$ -chains (van Urk, 2015, p. 40):
In a movement structure formed by Agree for $[\bar{A}]$ features, adjoin a node λf to the probing head.

- (63) $[_{DP} OP [_{NP}]] \lambda f_{cf} \dots [f_{cf}(NP)]$

In the $\bar{A}$ -chain depicted in (63), a counterpart operation of Trace Conversion replaces the lower OP with a choice function variable (e.g. Quantificational Copy Conversion proposed by van Urk, 2015). Besides the conversion operation, the higher OP is converted into a quantifier over choice functions and its NP complement is deleted.²² This means that, in $\bar{A}$ -chains, the non-operator part of the moved phrase must reconstruct.

A typical instance of $\bar{A}$ -chains is *wh*-movement. Assuming that *wh*-phrases are semantically *wh*-operators, the *wh*-question in (64) would have a simplified representation in (65), in which the *wh*-operator and its NP complement are disassociated.²³ At the head of the chain, everything but the *wh*-operator is deleted. Meanwhile, the lower copy of the *wh*-operator is converted into a choice function variable. The bound variable interpretation indicates that the complement of *which* is interpreted via its lower copy.

- (64) Which picture of herself_k does no woman_k like?

- (65) $[\text{which } [\text{picture of herself}_k]] \lambda f_{cf} \dots \text{no woman}_k \dots [f_{cf}(\text{picture of herself}_k)]$

In the *lian...dou* construction, movement of DP foci is the result of the $[\phi/\text{EVEN}]$ probe matching with a fully matching goal. Since both features are agreed, the interpretation rules in (60) and (62) can be potentially applied, resulting in the A- and $\bar{A}$ -chains in (66). The A-chain and the $\bar{A}$ -chain only differ in that the latter forces the fronted DP to reconstruct. However, the data presented in Sect. 3 suggests that $\bar{A}$ -chains are not possible for DP foci. If they were, DP foci should show reconstruction effects. In the following, I will argue that $\bar{A}$ -chains of DP foci are ruled out by economy.

- (66) a. A-chains: $[\text{lian-DP}] \lambda x_e \text{ dou} \dots [x_e]$
b. $\bar{A}$ -chains: $[\text{lian-}\bar{DP}] \lambda f_{cf} \text{ dou} \dots [f_{cf}(DP)]$

far more restricted than what syntactic reconstruction predicts, I assume reconstruction effects in A-chains can be attributed to semantics (Cresti, 1995; Rullmann, 1995; Lechner, 1998, 2013; Sternefeld, 2001; Ruys, 2015; Poole and Keine, 2024). This issue is irrelevant to the case here because A-chains in Mandarin Chinese do not reconstruct.

One reviewer suggests an alternative analysis in which reconstruction effects of fronted VPs and CPs are due to semantic reconstruction. I do not adopt this approach because it faces challenges that are not easy to solve. In the literature (see references above), semantic reconstruction is generally available in A-movement of DPs. However, to capture the asymmetry in reconstruction between DP foci and VP/CP foci in the *lian...dou* construction, semantic reconstruction would need to be blocked in A-movement of DP foci. I do not know how semantic reconstruction would enter into the economy calculation. It remains unclear to me what factors, if any, could block semantic reconstruction in cases involving DP foci.

²²It is beyond the scope of this paper to discuss how exactly an operator should be made computable with a predicate of choice functions but see Sauerland (1998) and Abels and Martí (2010) for possible solutions.

²³The exact computation of constituent-questions is not crucial for the purpose here. First, abstraction over choice functions is motivated independently from constituent-question semantics (Reinhart, 1997; Winter, 1997; Ruys, 2000). Second, most influential theories of constituent-question semantics are compatible with my main claim that the non-operator part of the moved phrase must reconstruct (Engdahl, 1980, 1986; Reinhart, 1997; Romero, 1998; Beck and Kim, 2006; Beck, 2006; Cable, 2007; Kotek, 2014; Heim, 2019).

6.2 Economy and costs of movement

In the generative theory of grammar, it has been a standard assumption that the principles of grammar should include principles of economy (for a recent overview of economy considerations in syntax and semantics, see Bošković and Messick, 2017). One formulation of the principles of economy is given in (67).

- (67) (Chomsky, 1995, p. 184)
 “If the derivation converges without the application of some operation, then that application is disallowed.”

In this paper, I adopt a more restrictive view that only representations with essentially the same semantic interpretation can be compared for economy of derivation (Golan, 1993; Fox, 1995, 2000; Reinhart, 2006). Under this view, when movement takes place for the same effect (including morphosyntactic and semantic effects) and several derivational options are available, movement takes place in the most economical way.

One manifestation of this economy effect on movement is that, all else being equal, a shorter movement chain is preferred. Example (68a) illustrates a typical superiority violation in English, where the lower rather than the higher *wh*-phrase moves. As shown in (68b) and (69b), moving either *who* or *what* yields the same interrogative interpretation. However, since the derivation in (68a) involves a longer movement chain than that in (69a), the latter is preferred by economy.

- (68) (Reinhart, 2006, p. 27)
 a. */? What_i did who buy t_i?
 b. For which $\langle x, y \rangle$, x bought y.

- (69) (Reinhart, 2006, p. 27)
 a. Who_i t_i bought what?
 b. For which $\langle x, y \rangle$, x bought y.

Notably, example (70a), where the derivation in (68a) happens in the embedded clause, is much better. Golan (1993) argues that example (70a) is grammatical because it yields an interpretation that is not available with the shorter chain in (71a). (70a) is acceptable only if the embedded *who* takes matrix scope, as illustrated in (70b). In contrast, (71a) only allows interpretations where the embedded *who* takes embedded scope, as shown in (71c and d). Thus, there is no alternative, more economical derivation that can achieve the interrogative interpretation in (70b). This suggests that the superiority effect is relativised to economy considerations, rather than being tied to some absolute syntactic constraints (e.g., the Superiority Condition by Chomsky, 1973).

- (70) (Reinhart, 2006, p. 26)
 a. Who_i t_i knows what_k who bought t_k?
 b. For which $\langle x, y \rangle$, x knows what y bought
 (e.g., *Lucie* knows what *Max* bought).
 c. *For which $\langle x, z \rangle$, x knows who bought z
 (e.g., *Lucie* knows who bought *a car*).

- d. *For which $\langle x \rangle$, x knows who bought what
(e.g., *Lucie* knows who bought what).
- (71) (Reinhart, 2006, p. 26)
- a. Who_i t_i knows who t_k bought what?
 - b. *For which $\langle x, y \rangle$, x knows what y bought
(e.g., *Lucie* knows what *Max* bought).
 - c. For which $\langle x, z \rangle$, x knows who bought z
(e.g., *Lucie* knows who bought *a car*).
 - d. For which $\langle x \rangle$, x knows who bought what
(e.g., *Lucie* knows who bought what).

I argue that composite probing in the *lian...dou* construction instantiates another economy effect on movement. Namely, when the $[\phi/\text{EVEN}]$ probe potentially triggers A- or $\bar{A}$ -chains to express the same EVEN interpretation, economy prefers A-chains because A-chains involve fewer derivational steps. As repeated in (72a), the A-chain only require Trace Conversion. In contrast, in the $\bar{A}$ -chains repeated in (71b), the derivation involves not only a comparable conversion operation but an additional deletion operation targeting the higher copy. I argue that the additional deletion operation is precisely what makes $\bar{A}$ -chains more costly. Thus, though A- and $\bar{A}$ -chains are equal in expressing the EVEN interpretation, A-chains are less costly.

- (72) a. A-chains: $[\text{lian-DP}] \lambda x_e \text{ dou} \dots [x_e]$ (preferred by economy)
 b. $\bar{A}$ -chains: $[\text{lian-DP}] \lambda f_{cf} \text{ dou} \dots [f_{cf}(\text{DP})]$

It should be noted that the specific number of derivational steps may vary depending on how one conceives the semantic derivation of A- and $\bar{A}$ -chains. However, it does not undermine the overall argument that A-chains are less costly in that they involve fewer derivational steps than $\bar{A}$ -chains.²⁴

Fox (2002) proposes that Trace Conversion consists of two operations: Variable Insertion and Determiner Replacement. In the economy calculus assumed here, I treat Trace Conversion as a single derivational step, abstracting away from its internal operations. One may take an alternative view that Variable Insertion and Determiner Replacement are two independent derivational steps. However, under this view, the counterpart of Trace Conversion in $\bar{A}$ -chains would also comprise (at least) two independent operations. At the same time, $\bar{A}$ -chains still require an extra deletion operation targeting the higher copy, making them more costly than A-chains.

An alternative derivation of A-chains is the Wholesale Late Merger mechanism proposed by Takahashi and Hulse (2009). Takahashi and Hulse (2009) claim that, while in A-chains, the NP restrictor of a determiner can merge with the determiner at the head of the chain, this kind of countercyclical merger is not possible for $\bar{A}$ -chains. Consequently, $\bar{A}$ -chains must reconstruct but A-chains do not have to. However, even if one adopts Takahashi and Hulse's proposal, s/he will arrive at the same conclusion

²⁴I assume that individual variables and choice function variables are the results of conversion operations with equal costs. If there are concerns, it is conceivable that conversion to a higher-type expression is more costly. This means that a choice function variable (type $\langle et, e \rangle$) incurs more costs than an individual-type variable (type $\langle e \rangle$).

that A-chains involve fewer derivational steps than $\bar{A}$ -chains. In a model with Wholesale Late Merger, A-chains require a modified version of Trace Conversion, which applies to copies of the determiner. $\bar{A}$ -chains still require a comparable conversion operation plus distributed deletion. Therefore, A-chains are generally less costly than $\bar{A}$ -chains, regardless of whether Wholesale Late Merger is allowed.

6.3 Deriving the A/ $\bar{A}$ distinction between DPs and non-DPs

In Sect. 4, one of the arguments for $\bar{A}$ -movement of VP and CP foci is the claim that these categories generally cannot undergo A-movement. In this section, I will provide a semantic account for this claim.

For VP and CP foci, the interpretation rule for A-chains cannot be applied because their semantic types are not compatible with individual-type variables. Therefore, there is no alternative, more economical derivations for VP and CP foci, resulting in $\bar{A}$ -movement of these foci.²⁵

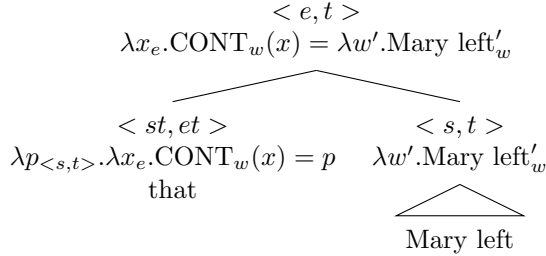
VPs are traditionally treated as predicates of type $\langle e, t \rangle$. Alternatively, by assuming the VP-internal Subject Hypothesis or the *v*P projection, VPs and their extended projections denote propositions of type $\langle s, t \rangle$.²⁶ Nevertheless, what is crucial is that the semantic type of VPs is not compatible with an individual variable, as shown in (73). Therefore, movement of VP foci cannot be translated into an A-chain.

(73) *A-chains: [*lian*-VP $_{\langle e, t \rangle}$] λx_e *dou*... [x_e] or [*lian*-VP $_{\langle s, t \rangle}$] λx_e *dou*... [x_e]

As for CPs, I adopt the predicate approach to the semantics of CPs (Kratzer, 2006; Moulton, 2009, 2015; Elliott, 2016; Bochnak and Hanink, 2022). Under this approach, finite CPs are predicates of propositional content (type $\langle e, t \rangle$). The complementiser C selects a proposition $\langle s, t \rangle$ and identifies this proposition as the content of a set of individuals, as in (74). A finite CP would have the simplified LF structure in (75).

(74) $\llbracket C \rrbracket = \lambda p_{\langle s, t \rangle} . \lambda x_e . \text{CONT}_w(x) = p$

(75) ... that Mary left



The ban on A-chains of CPs follows from the same rationale as that of VPs. Namely, the semantic type of CPs is not compatible with individual variables, as in (76).

²⁵The precise semantic representations of $\bar{A}$ -chains of VPs and CPs are beyond the scope of this paper. One potential direction to pursue is that $\bar{A}$ -movement of VPs and CPs also involve abstraction over choice function variables. However, choice functions are motivated by phenomena restricted to nominals (Reinhart, 1997; Winter, 1997; Kratzer, 1998; Schwarz, 2011). Therefore, extending existing theories of choice functions to account for $\bar{A}$ -movement of VPs and CPs is a non-trivial task. I will leave it aside here.

²⁶Once taking tense and aspect into consideration, VPs denote predicates of time $\langle i, st \rangle$ or of event $\langle v, st \rangle$.

(76) *A-chains: [*lian*-CP_{< e, t >}] λx_e *dou*... [*x_e*]

7 Conclusion and outlook

In this paper, I showed that in the *lian*...*dou* construction, the composite [ϕ /EVEN] probe triggers A-movement of DP foci but $\bar{A}$ -movement of VP and CP foci. This finding provides further empirical confirmation for Lohninger’s (2025a) generalisation that although [A] and [$\bar{A}$] features may form a composite probe, the movement chain they trigger is either A- or $\bar{A}$ -chains.

The main question this paper aims to answer is what determines the triggering feature in the composite [ϕ /EVEN] probe. I argue that while the [ϕ /EVEN] probe potentially triggers A- or $\bar{A}$ -chains of DP foci, economy prefers the former. A- and $\bar{A}$ -chains differ in their interpretation rules: A-chains abstract over individuals while $\bar{A}$ -chains abstract over choice functions. What is relevant to economy considerations is that the interpretation rule for A-chains involves fewer derivational steps. Therefore, all else being equal, economy prefers A-chains over $\bar{A}$ -chains. Meanwhile, the interpretation rule for A-chains cannot be applied to VP and CP foci due to type mismatch. Consequently, while DP foci undergo A-movement due to economy, VP and CP foci undergo $\bar{A}$ -movement.

Lohninger (2025a) suggests two possible determining factors of the movement trigger in composite [A/ $\bar{A}$] probes. The first candidate is the strength of features. Lohninger (2025a) assumes that agreeing features can be merged as strong or weak features (Chomsky, 1995). In a composite [A/ $\bar{A}$] probe, the strong feature triggers movement due to its EPP property, whereas the weak feature is agreed as a free rider. There are two main problems with this approach. The first problem is conceptual. This approach does not answer the question; it merely restates it in a different form. The strength of features is highly stipulative and not well-defined in the literature. It is unclear whether features being strong or weak is universal cross-linguistically, or should be relativised to specific languages. The question of what determines the triggering feature in composite [A/ $\bar{A}$] probes thus becomes a question of what determines the strength of features. The second problem is empirical. Recall that, in the *lian*...*dou* construction, the [ϕ] feature triggers A-movement of DP foci. This suggests that the [ϕ] feature should be a strong EPP feature in the [ϕ /EVEN] probe. However, this wrongly predicts that the [ϕ] feature can trigger movement of a plain [ϕ] goal when it probes independently. To avoid such overgeneralisation, we must stipulate that, within a single configuration, the [ϕ] feature is sometimes strong, and sometimes weak. However, I do not believe this is a satisfactory solution.

Lohninger’s (2025a) another idea, which aligns with the main claim of this paper, is that an independent principle determines that A-movement takes precedence over $\bar{A}$ -movement in composite A/ $\bar{A}$ constructions. This idea has been implemented differently in our respective analyses. In my analysis, I propose that economy prefers A-movement over $\bar{A}$ -movement. Lohninger (2025a) adopts the hierarchy of movement operations by Abels (2007) and argues that A-movement is preferred because it is timed before $\bar{A}$ -movement on the hierarchy. In other words, if A-movement is available, it obligatorily applies. Notably, the typological findings by Lohninger (2025a) suggest

that the preference for A-chains is cross-linguistically universal in composite A/ $\bar{A}$ constructions. According to Lohninger (2025a), the only instance of A/ $\bar{A}$ probes triggering $\bar{A}$ -movement is long-distance agreement constructions. By contrast, instances of [A/ $\bar{A}$] probes triggering A-movement are widely attested in cross-clausal A-movement, promotion to pivot, topicalization, focalisation and (noncanonical) passive constructions in different languages. If A-movement is indeed prioritised—whether due to economy or the hierarchy of movement operations—it is unsurprising that the preference for A-chains correlates with a higher statistical tendency for A-movement in the typology of composite A/ $\bar{A}$ constructions. Another difference is that Lohninger’s (2025a) implementation makes a stronger prediction: there should be no $\bar{A}$ -movement in composite A/ $\bar{A}$ constructions (see also Lohninger, 2024, 2025b). Under this view, an alternative analysis of long-distance agreement constructions is required (e.g., Preminger, 2009). By contrast, my economy-based analysis predicts that $\bar{A}$ -movement in composite A/ $\bar{A}$ constructions would be exceptions to the economy preference. At present, it is unclear to me what the violable conditions are, and the available language sample is too limited to test these predictions. I will leave these for future research.

Abbreviations

1 = first person, 3 = third person, ADJ = adjective, ADV = adverb(ial), ASP = aspect marker, CLF = classifier, COMP = complementizer, COP = copula, EXP = experience, NEG = negation, NF = non-finite, OV = object voice, P = preposition, PASS = passive, PRF = perfect, POSS = possessive, PRT = particles, REL = relative clause marker, SG = singular, SFP = sentence-final particle, TOP = topic

References

- Abels, K. 2007. Towards a restrictive theory of (remnant) movement! *Linguistic Variation Yearbook* 7: 53–120. <https://doi.org/10.1075/livy.7.04abe> .
- Abels, K. 2012. *Phases: An Essay on Cyclicity in Syntax*. Berlin: Walter de Gruyter.
- Abels, K. and L. Martí. 2010. A unified approach to split scope. *Natural Language Semantics* 18(4): 435–470. <https://doi.org/10.1007/s11050-010-9060-8> .
- Aldridge, E.C. 2004. *Ergativity and Word Order in Austronesian Languages*. Ph. D. thesis, Cornell University.
- Alrenga, P. 2005. A sentential subject asymmetry in English and its implications for complement selection. *Syntax* 8(3): 175–207. <https://doi.org/10.1111/j.1467-9612.2005.00078.x> .
- Aoyagi, H. 1998. *On the Nature of Particles in Japanese and Its Theoretical Implications*. Ph. D. thesis, University of Southern California.
- Badan, L. 2008. Preposed object and low periphery in Mandarin Chinese. In *Still Working CISCL Working Papers on Language and Cognition*. <http://mitwpl.mit>.

edu/.

- Badan, L. 2012. 'Lian.....dou' he 'lian.....ye' de jufa yuyi ji yuyong chayi (On the Discrimination of 'Lian.....dou.....'and 'Lian.....ye.....'). *Hanyu Xuexi (Chinese Language Learning)* 3: 50–58 .
- Badan, L. and F. Del Gobbo. 2015. The even-construction and the low periphery in Mandarin Chinese, In *The Cartography of Chinese Syntax*, ed. Tsai, W.T.D., 33–74. New York: Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780190210687.003.0002>.
- Bárány, A. 2023. Interaction, satisfaction, and (a)symmetric object agreement. Talk at NELS53, Göttingen, Germany.
- Barss, A. 1986. *Chains and Anaphoric Dependence: On Reconstruction and Its Implications*. Ph. D. thesis, Massachusetts Institute of Technology.
- Barss, A. 2001. Syntactic Reconstruction Effects, In *The Handbook of Contemporary Syntactic Theory* (1 ed.), eds. Baltin, M. and C. Collins, 670–696. Wiley.
- Bayer, J. 1996. *Directionality and Logical Form: On the Scope of Focusing Particles and Wh-in-Situ*. Number v. 34 in Studies in Natural Language and Linguistic Theory. Dordrecht ; Boston: Kluwer Academic.
- Beck, S. 2006. Intervention Effects Follow from Focus Interpretation. *Natural Language Semantics* 14(1): 1–56. <https://doi.org/10.1007/s11050-005-4532-y> .
- Beck, S. and S.S. Kim. 2006. Intervention Effects in Alternative Questions. *The Journal of Comparative Germanic Linguistics* 9(3): 165–208. <https://doi.org/10.1007/s10828-006-9005-2> .
- Bochnak, M.R. and E.A. Hanink. 2022. Clausal embedding in Washo: Complementation vs. modification. *Natural Language & Linguistic Theory* 40(4): 979–1022. <https://doi.org/10.1007/s11049-021-09532-z> .
- Bošković, Ž. and T. Messick. 2017. Derivational Economy in Syntax and Semantics, *Oxford Research Encyclopedia of Linguistics*. Oxford University Press. <https://doi.org/10.1093/acrefore/9780199384655.013.130>.
- Bossi, M. and M. Diercks. 2019. V1 in Kipsigis: Head movement and discourse-based scrambling. *Glossa: a journal of general linguistics* 4(1). <https://doi.org/10.5334/gjgl.246> .
- Branan, K. and M.Y. Erlewine. 2024. $\bar{A}$ -Probing for the Closest DP. *Linguistic Inquiry* 55(2): 375–401. <https://doi.org/10.1162/ling.a.00459> .
- Cable, S. 2007. *The Grammar of Q: Q-particles and the Nature of Wh-fronting, as Revealed by the Wh-questions of Tlingit*. Ph. D. thesis, Massachusetts Institute of

Technology.

- Cao, H. 2020. The Multi-Type of Movement in Lian... Dou Construction in Mandarin Chinese and Its Syntactic and Semantic Licenses. Master's thesis, University College London.
- Cao, H. 2024. One trigger, multiple movement types: An economy approach to the Chinese even-construction. In *The Proceedings of the 60th Meeting of the Chicago Linguistic Society (CLS 60)*, Chicago.
- Cao, H. 2025. *The Syntax of Focus Movement in The Mandarin Lian... Dou Construction: The Trigger, The Movement Paths and The Landing Site*. Ph. D. thesis, University College London, expected December, 2025.
- Chen, F. 2023. *Obscured Universality in Mandarin*. Ph. D. thesis, Massachusetts Institute of Technology.
- Chen, F. 2024. Generalized composite probing in Mandarin. Ms., Massachusetts Institute of Technology.
- Cheng, L.L.S. and L. Vicente. 2013. Verb doubling in Mandarin Chinese. *Journal of East Asian Linguistics* 22(1): 1–37. <https://doi.org/10.1007/s10831-012-9095-6>.
- Chomsky, N. 1973. Conditions on transformations, In *A Festschrift for Morris Halle*, eds. Anderson, S. and P. Kiparsky, 232–286. New York: Holt, Reinhart and Winston.
- Chomsky, N. 1977. On wh-movement, In *Formal Syntax*, eds. Culicover, P.W., T. Wasow, and A. Akmajian, 71–132. New York: Academic Press.
- Chomsky, N. 1981. *Lectures on Government and Binding*. Number 9 in Studies in Generative Grammar. Dordrecht: Foris Publications.
- Chomsky, N. 1993. A minimalist Program for Linguistic Theory, In *The View from Building 20 Essays in Linguistics in Honor of Sylvain Bromberger*, eds. Hale, K. and S.J. Keyser, 1–52. Cambridge, Mass: MIT Press.
- Chomsky, N. 1995. *The Minimalist Program*. Number 28 in Current Studies in Linguistics. Cambridge, Mass: MIT Press.
- Chomsky, N. 2001. Derivation by phase, In *Ken Hale: A Life in Nguage*, ed. Kenstowicz, M. ambridge, MA: MIT Press.
- Chomsky, N. 2007. Approaching UG from Below, In *Interfaces + Recursion = Language?*, eds. Sauerland, U. and H.M. Gärtner, 1–30. Berlin, München, Boston: Mouton de Gruyter. <https://doi.org/10.1515/9783110207552-001>.
- Chou, C.T.T. 2024. Satisfying the composite probe on the Topic head in Mandarin Chinese. *Studia Linguistica*: stul.12247. <https://doi.org/10.1111/stul.12247>.

- Cinque, G. 1999. *Adverbs and Functional Heads: A Cross-Linguistic Perspective*. Oxford Studies in Comparative Syntax. New York: Oxford University Press.
- Citko, B. 2008. Missing labels. *Lingua* 118(7): 907–944. <https://doi.org/10.1016/j.lingua.2008.01.001> .
- Constant, N. and C.C. Gu 2010. Mandarin ‘even’, ‘all’ and the trigger of focus movement. In *University of Pennsylvania Working Papers in Linguistics (Proceedings of the 33rd Annual Penn Linguistics Colloquium)*, Volume 16, pp. 21–30.
- Coon, J., N. Baier, and T. Levin. 2021. Mayan agent focus and the ergative extraction constraint: Facts and fictions revisited. *Language* 97(2): 269–332. <https://doi.org/10.1353/lan.2021.0019> .
- Coon, J. and A. Bale. 2014. The interaction of person and number in Mi’gmaq. *Nordlyd* 41(1): 85. <https://doi.org/10.7557/12.3235> .
- Coon, J. and S. Keine. 2021. Feature Gluttony. *Linguistic Inquiry* 52(4): 655–710. https://doi.org/10.1162/ling_a.00386 .
- Cresti, D. 1995. Extraction and reconstruction. *Natural Language Semantics* 3(1): 79–122. <https://doi.org/10.1007/BF01252885> .
- Crnič, L. 2011. *Getting Even*. Ph. D. thesis, Massachusetts Institute of Technology.
- Déprez, V.M. 1989. *On the Typology of Syntactic Positions and the Nature of Chains: Move [Alpha] to the Specifier of Functional Projections*. Ph. D. thesis, Massachusetts Institute of Technology.
- Djamouri, R. and W. Paul. 2017. A new approach to-zhe in Mandarin Chinese, *Studies in Japanese and Korean Historical and Theoretical Linguistics and Beyond*, 110–123. Leiden: Brill.
- Elbourne, P.D. 2005. *Situations and Individuals*. Number 41 in Current Studies in Linguistics. Cambridge, Mass: MIT Press.
- Elliott, P.D. 2016. Explaining DPs vs. CPs without syntax. In J. Kantarovich, T. Truong, and O. Xherija (Eds.), *Proceedings of Chicago Linguistic Society (CLS) 52*, Chicago, pp. 171–185. Chicago Linguistic Society.
- Engdahl, E. 1980. *The Syntax and Semantics of Questions in Swedish*. Ph. D. thesis, University of Massachusetts, Amherst.
- Engdahl, E. 1986. *Constituent Questions: The Syntax and Semantics of Questions with Special Reference to Swedish*. Studies in Linguistics and Philosophy. Dordrecht: D. Reidel Pub. Co.

- Erlewine, M.Y. 2018. Extraction and Licensing in Toba Batak. *Language* 94(3): 662–697. <https://doi.org/10.1353/lan.2018.0039>. 26630375 .
- Ernst, T. 2014a. Adverbial Adjuncts in Mandarin Chinese, In *The Handbook of Chinese Linguistics* (1 ed.), eds. Huang, C.T.J., Y.H.A. Li, and A. Simpson, 49–72. Hoboken, USA: John Wiley & Sons. <https://doi.org/10.1002/9781118584552.ch3>.
- Ernst, T. 2014b. The syntax of adverbs, *The Routledge Handbook of Syntax*, 108–130. Routledge.
- Ernst, T. and C. Wang. 1995. Object preposing in Mandarin Chinese. *Journal of East Asian Linguistics* 4(3): 235–260. <https://doi.org/10.1007/BF01731510> .
- Feng, Y. and H. Pan. 2023. Chinese Dou Quantification. *Oxford Research Encyclopedia of Linguistics*. <https://doi.org/10.1093/acrefore/9780199384655.013.890> .
- Fox, D. 1995. Economy and scope. *Natural Language Semantics* 3(3): 283–341. <https://doi.org/10.1007/BF01248820> .
- Fox, D. 1999. Reconstruction, Binding Theory, and the Interpretation of Chains. *Linguistic Inquiry* 30(2): 157–196. <https://doi.org/10.1162/002438999554020> .
- Fox, D. 2000. *Economy and Semantic Interpretation*. Number 35 in Linguistic Inquiry Monographs. Cambridge, Mass: MIT press.
- Fox, D. 2002. Antecedent-Contained Deletion and the Copy Theory of Movement. *Linguistic Inquiry* 33(1): 63–96. <https://doi.org/10.1162/002438902317382189> .
- Fox, D. 2003. On Logical Form, In *Minimalist Syntax* (1 ed.), ed. Hendrick, R., 82–123. Oxford, UK: Blackwell Publishing Ltd. <https://doi.org/10.1002/9780470758342.ch2>.
- Georgi, D. 2014. *Opaque Interactions of Merge and Agree: On the Nature and Order of Elementary Operations*. Ph. D. thesis, Universität Leipzig.
- Golan, Y. 1993. Node crossing economy, superiority and D-linking. Ms. , Tel Aviv University.
- Heim, I. 2019. Functional readings without type-shifted noun phrases, In *Reconstruction Effects in Relative Clauses*, eds. Krifka, M. and M. Schenner, 283–302. Berlin, Boston: De Gruyter. <https://doi.org/10.1515/9783050095158-009>.
- Heycock, C. 1995. Asymmetries in reconstruction. *Linguistic inquiry* 26(4): 547–570. [jstor:4178914](https://www.jstor.org/stable/4178914) .
- Hornstein, N. 1995. *Logical Form: From GB to Minimalism*. Oxford: Blackwell.

- Hsu, Y.Y. 2008. The Sentence-Internal Topic and Focus in Chinese. In *Annual Meeting of the Berkeley Linguistics Society*, Volume 34, pp. 437–448.
- Hsu, Y.Y. 2013. *The Interaction of Information Structure and Syntactic Representation in Chinese*. Ph. D. thesis, Indiana University.
- Huang, C.T.J. 1993. Reconstruction and the structure of VP: Some theoretical consequences. *Linguistic inquiry* 24(1): 103–138. jstor:4178803 .
- Karttunen, L. and S. Peters. 1979. Conventional implicature, In *Presupposition*, eds. David, A.D. and C.K. Oh, Volume 11 of *Syntax and Semantics*, 1–56. New York: Academic Press.
- Keine, S. 2019. Selective Opacity. *Linguistic Inquiry* 50(1): 13–62. https://doi.org/10.1162/ling_a_00299 .
- Kobele, G.M. 2012. Deriving Reconstruction Asymmetries, In *Local Modelling of Non-Local Dependencies in Syntax*, eds. Alexiadou, A., T. Kiss, and G. Müller, 477–500. Berlin, Boston: DE GRUYTER. <https://doi.org/10.1515/9783110294774.477>.
- Koster, J. 1978. Why subject sentences don't exist, In *Recent Transformational Studies in European Languages*, ed. Keyser, S.J., 53–64. Cambridge, Mass: MIT Press.
- Kotek, H. 2014. *Composing Questions*. Ph. D. thesis, Massachusetts Institute of Technology.
- Kratzer, A. 1998. More Structural Analogies between Pronouns and Tenses. In *Proceedings of SALT 8*, pp. 92–109.
- Kratzer, A. 2006. Decomposing attitude verbs. Ms., University of Massachusetts at Amherst.
- Krifka, M. 1991. A Compositional Semantics for Multiple Focus Constructions. *Semantics and Linguistic Theory* 1: 127–158. <https://doi.org/10.3765/salt.v1i0.2492> .
- Kuo, C.M. 2003. *The Fine Structure of Negative Polarity Items in Chinese*. Ph. D. thesis, University of Southern California.
- Lebeaux, D. 2009. *Where Does Binding Theory Apply?* Number 50 in *Linguistic Inquiry Monographs*. Cambridge, Mass: MIT Press.
- Lechner, W. 1998. Two kinds of reconstruction. *Studia Linguistica* 52(3): 276–310. <https://doi.org/10.1111/1467-9582.00037> .
- Lechner, W. 2013. Diagnosing Covert Movement: The Duke of York and Reconstruction, In *Diagnosing Syntax*, eds. Cheng, L.L.S. and N. Corver, 158–189. Oxford: Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199602490.003>.

0008.

- Lee, P.P.I. 2019. *Focus Manifestation in Mandarin Chinese and Cantonese: A Comparative Perspective* (1 ed.). London: Routledge.
- Li, P. 2020. *Topics in the Mandarin Lian... Dou Construction: Its Syntax and Acquisition*. Ph. D. thesis, Boston University.
- Liao, H.C. 2011. *Alternatives and Exhaustification: Non-interrogative Uses of Chinese Wh- Words*. Ph. D. thesis, Harvard University.
- Lohninger, M. 2024. [A'/A] probing is just edgy A-movement. Invited talk at Stony Brook University.
- Lohninger, M. 2025a. *Patterns in Chaos: Composite A'/A Probes*. Ph. D. thesis, University of Vienna.
- Lohninger, M. 2025b. There is no composite A'/A movement. Poster at GLOW47.
- Lohninger, M., I. Kovač, and S. Wurmbrand. 2022. From Prolepsis to Hyperraising. *Philosophies* 7(2): 32. <https://doi.org/10.3390/philosophies7020032> .
- Lohninger, M. and K.F. Yip 2023. (In)dependence of features on composite probes. In S.-Y. Lam and S. Ozaki (Eds.), *Proceedings of the Fifty-Third Annual Meeting of the North East Linguistic Society (NELS)*, Volume 2, pp. 155–164.
- Longenbaugh, N. 2017. Composite A/A'-movement: Evidence from English tough-movement. Ms., Massachusetts Institute of Technology.
- Mahajan, A.K. 1990. *The A/A-bar Distinction and Movement Theory*. Ph. D. thesis, Massachusetts Institute of Technology.
- May, R.C. 1977. *The Grammar of Quantification*. Ph. D. thesis, Massachusetts Institute of Technology.
- Moulton, K. 2009. *Natural Selection and the Syntax of Clausal Complementation*. Ph. D. thesis, University of Massachusetts, Amherst.
- Moulton, K. 2013. Not Moving Clauses: Connectivity in Clausal Arguments. *Syntax* 16(3): 250–291. <https://doi.org/10.1111/synt.12007> .
- Moulton, K. 2015. CPs: Copies and Compositionality. *Linguistic Inquiry* 46(2): 305–342. https://doi.org/10.1162/LING_a-00183 .
- Neeleman, A. and K. Szendrői. 2004. Superman sentences. *Linguistic Inquiry*: 149–159. [jstor:4179266](https://www.jstor.org/stable/4179266) .

- Neeleman, A. and H. van de Koot. 2010. A Local Encoding of Syntactic Dependencies and Its Consequences for the Theory of Movement. *Syntax* 13(4): 331–372. <https://doi.org/10.1111/j.1467-9612.2010.00143.x> .
- Obata, M. and S.D. Epstein. 2011. Feature-Splitting Internal Merge: Improper Movement, Intervention, and the A/A' Distinction. *Syntax* 14(2): 122–147. <https://doi.org/10.1111/j.1467-9612.2010.00149.x> .
- Pan, V.J. 2017. Optional projections in the left-periphery in Mandarin Chinese, In *Studies on Syntactic Cartography*, ed. Si, F., 218–248. Beijing: Zhongguó shèhuì chubǎnshè [China Social Sciences Press].
- Paul, W. 2002. Sentence-internal topics in Mandarin Chinese: The case of object preposing. *Language and Linguistics* 3(4): 695–714 .
- Poole, E. 2023. Improper case. *Natural Language & Linguistic Theory* 41(1): 347–397. <https://doi.org/10.1007/s11049-022-09541-6> .
- Poole, E. 2024. (Im)possible Traces. *Linguistic Inquiry* 55(2): 287–326. https://doi.org/10.1162/ling_a_00467 .
- Poole, E. and S. Keine. 2024. Not all reconstruction effects are syntactic. *Natural Language & Linguistic Theory*. <https://doi.org/10.1007/s11049-023-09603-3> .
- Postal, P.M. 1971. *Cross-over Phenomena*. Transatlantic Series in Linguistics. New York: Holt, Rinehart and Winston.
- Preminger, O. 2009. Breaking Agreements: Distinguishing Agreement and Clitic Doubling by Their Failures. *Linguistic Inquiry* 40(4): 619–666. <https://doi.org/10.1162/ling.2009.40.4.619> .
- Preminger, O. 2011. *Agreement as a Fallible Operation*. Ph. D. thesis, Massachusetts Institute of Technology.
- Preminger, O. 2014. *Agreement and Its Failures*. Cambridge, Massachusetts ; London, England: The MIT Press.
- Qu, Y. 1994. *Object Noun Phrase Dislocation in Mandarin Chinese*. Ph. D. thesis, University of British Columbia.
- Reinhart, T. 1997. WH-IN-SITU in the framework of the minimalist program. *Natural Language Semantics* 6(1): 29–56. <https://doi.org/10.1023/A:1008240014550> .
- Reinhart, T. 2006. *Interface Strategies: Optimal and Costly Computations*. The MIT Press.
- Rizzi, L. 1990. *Relativized Minimality*. Cambridge, MA: The MIT Press.

- Rizzi, L. 1997. The Fine Structure of the Left Periphery, In *Elements of Grammar*, ed. Haegeman, L., 281–337. Dordrecht: Springer Netherlands. https://doi.org/10.1007/978-94-011-5420-8_7.
- Rizzi, L. 2004. *The Structure of CP and IP: The Cartography of Syntactic Structures, Volume 2*. Oxford: Oxford University Press.
- Romero, M. 1998. *Focus and Reconstruction Effects in Wh-Phrases*. Ph. D. thesis, University of Massachusetts, Amherst.
- Rooth, M. 1985. *Association with Focus*. Ph. D. thesis, University of Massachusetts, Amherst.
- Rooth, M. 1992. A theory of focus interpretation. *Natural Language Semantics* 1(1): 75–116. <https://doi.org/10.1007/BF02342617> .
- Rullmann, H. 1995. *Maximality in the Semantics of Wh-Constructions*. Ph. D. thesis, University of Massachusetts, Amherst.
- Ruys, E.G. 2000. Weak Crossover as a Scope Phenomenon. *Linguistic Inquiry* 31(3): 513–539. <https://doi.org/10.1162/002438900554424> .
- Ruys, E.G. 2015. A Minimalist Condition on Semantic Reconstruction. *Linguistic Inquiry* 46(3): 453–488. https://doi.org/10.1162/LING_a_00189 .
- Safir, K. 2019. The A/ $\bar{A}$ Distinction as an Epiphenomenon. *Linguistic Inquiry* 50(2): 285–336. https://doi.org/10.1162/ling_a.00305 .
- Sauerland, U. 1998. *The Meaning of Chains*. Ph. D. thesis, Massachusetts Institute of Technology.
- Sauerland, U. 2004. The Interpretation of Traces. *Natural Language Semantics* 12(1): 63–127. <https://doi.org/10.1023/B:NALS.0000011201.91994.4f> .
- Sauerland, U. and P. Elbourne. 2002. Total Reconstruction, PF Movement, and Derivational Order. *Linguistic Inquiry* 33(2): 283–319. <https://doi.org/10.1162/002438902317406722> .
- Schwarz, B. 2011. Long Distance Indefinites and Choice Functions. *Language and Linguistics Compass* 5(12): 880–897. <https://doi.org/10.1111/j.1749-818X.2011.00315.x> .
- Scott, T. 2021. Formalizing two types of mixed A/ $\bar{A}$ movement. Ms., University of California Berkeley.
- Shyu, S.i. 1995. *The Syntax of Focus and Topic in Mandarin Chinese*. Ph. D. thesis, University of Southern California.

- Shyu, S.i. 2001. Remarks on object movement in Mandarin SOV order. *Language and Linguistics* 2(1): 93–124 .
- Shyu, S.i. 2016. Minimizers and EVEN. *Linguistics* 54(6): 1355–1395. <https://doi.org/10.1515/ling-2016-0031> .
- Shyu, S.i. 2018. The scope of even: Evidence from Mandarin Chinese. *Language and Linguistics* 19(1): 156–195. <https://doi.org/10.1075/lali.00006.shy> .
- Simpson, A. 2014. Prosody and Syntax, In *The Handbook of Chinese Linguistics* (1 ed.), eds. Huang, C.T.J., Y.H.A. Li, and A. Simpson, 465–491. Wiley. <https://doi.org/10.1002/9781118584552.ch18>.
- Sportiche, D. 2017. Reconstruction, Binding, and Scope, In *The Wiley Blackwell Companion to Syntax* (2 ed.), eds. Everaert, M. and H.C. Riemsdijk, 3569–3626. Oxford: Wiley. <https://doi.org/10.1002/9781118358733.wbsyncom002>.
- Sternefeld, W. 2001. Semantic vs. syntactic reconstruction, In *Linguistic Form and Its Computation*, eds. Kamp, H., A. Rossdeutscher, and C. Rohrer, Number 1 in Studies in Computational Linguistics, 145–182. Stanford, Calif: CSLI Publications.
- Sybesma, R. 2021. Voice and Little v and VO–OV Word-Order Variation in Chinese Languages. *Syntax* 24(1): 44–77. <https://doi.org/10.1111/synt.12211> .
- Takahashi, S. 2010. The hidden side of clausal complements. *Natural Language & Linguistic Theory* 28(2): 343–380. <https://doi.org/10.1007/s11049-010-9091-3> .
- Takahashi, S. and S. Hulsey. 2009. Wholesale Late Merger: Beyond the A/ $\bar{A}$ Distinction. *Linguistic Inquiry* 40(3): 387–426. <https://doi.org/10.1162/ling.2009.40.3.387> .
- Takano, Y. 1995. Predicate fronting and internal subjects. *Linguistic Inquiry* 26(2): 327–340. jstor:4178899 .
- van Riemsdijk, H. and E. Williams. 1981. NP-strcuture. *The Linguistic Review* 1(2). <https://doi.org/10.1515/tlir.1981.1.2.171> .
- van Urk, C. 2015. *A Uniform Syntax for Phrasal Movement: A Case Study of Dinka Bor*. Ph. D. thesis, Massachusetts Institute of Technology.
- Wang, Y.I. and W.w.R. Liao. 2021. Against the Low Periphery as Phrase Markers: Two Types of Contrastive Dislocations in Mandarin Chinese*. *Studia Linguistica* 75(1): 97–127. <https://doi.org/10.1111/stul.12151> .
- Williams, E. 2003. *Representation Theory*. Cambridge, Mass: MIT Press.
- Winter, Y. 1997. Choice Functions and the Scopal Semantics of Indefinites. *Linguistics and Philosophy* 20(4): 399–467. jstor:25001672 .

- Xiang, M. 2008. Plurality, maximality and scalar inferences: A case study of Mandarin Dou. *Journal of East Asian Linguistics* 17(3): 227–245. <https://doi.org/10.1007/s10831-008-9025-9> .
- Xiang, Y. 2020. Function Alternations of the Mandarin Particle Dou: Distributor, Free Choice Licensor, and ‘Even’. *Journal of Semantics* 37(2): 171–217. <https://doi.org/10.1093/jos/ffz018> .
- Yin, B. 2023. Verb Doubling in Mandarin Chinese is Base-generated. Ms., UCL (University College London).